

# Markups and Misallocation in Banking

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## Abstract

We study how uneven bank pricing power distorts the joint allocation of deposits and loans across banks. We develop a structural model tailored to the two-sided environment and show that the welfare loss from misallocation can be summarized by the weighted dispersion of deposit and loan markups across banks, together with a third component that captures how the two sides interact. Applying the framework to U.S. banking data, we find that markup-driven misallocation reduced welfare by about 0.2 to 0.6 percent of total assets per year in the 1980s and 1990s, but declined sharply in recent years. The deposit side accounts for most of the measured loss, and markup dispersions on the deposit and loan sides generally reinforce each other in generating welfare losses. We then use the framework to study bank mergers and find that mergers on average increase the acquirers' contribution to misallocation, especially for deals in the 1990s. Finally, we construct an *ex ante* measure of predicted welfare loss from misallocation for each deal based on an ownership counterfactual, and confirm that deals with larger predicted welfare losses *ex ante* are associated with larger *ex post* increases in misallocation.

**JEL Classification:** G21, L13, D61, G34

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# 1 Introduction

The allocation of banking activity is important for the economy as banks decide where a large stock of liquid savings is turned into credit. A dollar of deposits can support very different social value depending on which bank collects it, and a dollar of loans can have very different cost depending on which bank supplies it. This is exactly the central question in the misallocation literature: aggregate output depends not only on how much capital and labor the economy has, but also on whether resources flow to the producers with the highest marginal returns (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009). This allocative margin matter empirically because banking is both large and uneven. U.S. banks hold trillions of dollars of deposits and loans, but households and firms do not face a frictionless national market for bank services.

This paper studies how market power distorts the allocation of deposits and loans across banks. Deposits are misallocated when they are not collected by the banks that value them most at the margin, and loans are misallocated when they are not supplied by the banks that can produce them at the lowest marginal cost. In an efficient banking system, deposits and loans flow across banks until these marginal gaps are closed. *Uneven* pricing power prevents the free flow, as high-markup banks attract too little activity in equilibrium relative to the efficient allocation and low-markup banks become too large.

Banking makes this problem especially interesting because deposits and loans are jointly produced. The two sides are linked through demand, as a strong deposit franchise can attract borrowers and a strong lending franchise can attract depositors. They are also connected through costs, since deposits fund loans and can lower the marginal cost of lending. As a result, misallocation in banking cannot be measured from deposits or loans in isolation. It depends on how both sides of the balance sheet are allocated across banks.

The paper proceeds in three steps. First, we develop a framework tailored to two-sided intermediation that maps the dispersion of banks' markups into welfare losses. A planner resets deposit and loan rates while holding aggregate deposits and aggregate loans fixed, thereby reallocating quantities across banks to maximize the total surplus of banks, depositors, and borrowers. This isolates the allocative problem within banking from the separate question of how large the banking sector should be. At the social optimum, the planner equalizes markups across banks on each side. This motivates our definition of the *social wedge* as the gap between a bank's markup and the common planner benchmark on that side. The resulting welfare loss admits a local decomposition into three components, one from

deposits, one from loans, and one from two-sided interactions. Each component depends on the weighted dispersion of these social wedges across banks, where the weights are given by the curvature of total surplus and capture how much quantity reallocation is required to reach the social optimum. We refer to this mechanism as *markup-driven misallocation*.

Because deposits and loans are linked, banks do not set markups on the two sides independently. A change in deposit pricing affects not only the quantity demand of deposits, but also that of loans, and the same is true in reverse. Thus, markups are jointly determined across the two sides of the balance sheet, and so are the corresponding social wedges. The two-sided structure also matters for how wedge dispersion translates into welfare loss. A distortion in deposit pricing can change the welfare cost of loan distortions, and vice versa, because reallocating deposits across banks also changes loan-side surplus and balance-sheet costs. Therefore, the welfare loss includes an interaction term that captures whether deposit and loan wedges amplify or dampen each other in generating welfare losses from misallocation.

Second, we take the framework to the data. We estimate a two-sided demand system for deposits and loans using U.S. bank data and recover the sufficient statistics needed to measure markup-driven misallocation. The demand side follows the structural banking literature in allowing for differentiated banks and heterogeneous price sensitivity (Egan et al., 2017; Wang et al., 2022). The key extension is that demand on each side depends on the bank’s presence on the other side. This reduced-form linkage captures relationship banking, one-stop banking, trust, and other cross-side complementarities. The estimated demand system then delivers bank-specific elasticities, which in turn generate the dispersion in markups that drives the welfare loss.

The welfare loss estimates imply that markup-driven misallocation in banking is economically meaningful. In the 1980s and 1990s, the implied welfare loss is about 0.2 to 0.6 percent of total bank assets, or about \$3.7 to \$22.3 billion in 2019 dollars per year, with substantially smaller losses in recent years. The deposit side accounts for most of the measured welfare loss, consistent with the much larger cross-sectional dispersion in deposit markups than in loan markups. This difference is in turn driven by the greater dispersion in depositors’ price sensitivity relative to borrowers’. The interaction component is positive and sizable in most years, which implies that deposit-side and loan-side distortions usually reinforce each other in generating welfare loss. At the same time, the counterfactuals show that cross-side demand links compress markup dispersion within each side enough to lower total welfare loss on net, even though they raise the contribution of the interaction component. Taken together, these

results show that a one-sided view of banking would have missed an important part of the allocative problem.

Finally, we apply the framework to bank mergers and acquisitions. Using a stacked event-study design, we find that mergers tend to move acquiring banks' markups further away from the planner's benchmark after the deal, so the misallocation-driven welfare loss contributed by these banks rises on average. The effect is strongest on the deposit side and is especially pronounced for mergers in the 1990s. To highlight the heterogeneity across deals, we then construct an *ex ante* measure of predicted changes in welfare loss for each merger deal. The exercise is based on a simple ownership counterfactual in which we keep fixed the pre-merger demand system and observed quantities, and change only the ownership structure so that the acquirer and target jointly internalize the profits of both banks. Intuitively, business diverted from one merging party by the other is no longer treated as a loss to a rival, which changes their pricing incentives and the markups implied by the model. We then map the resulting change in markup dispersion into the deal-level changes in welfare loss from misallocation. We find that most deals are close to neutral but the distribution has a meaningful right tail, so a smaller set of deals materially worsens misallocation and keeps the average effect positive. Consistent with this interpretation, acquirers involved in deals with larger *ex ante* predicted losses also show larger *ex post* increases in absolute social wedges and in welfare loss from misallocation.

Taken together, these results show that bank market power matters not only for average spreads and monetary policy transmission, as emphasized in the literature, but also for how intermediation activity is allocated across institutions. More broadly, they suggest that evaluating competition in banking requires moving beyond average rates and concentration measures to the cross-sectional dispersion of pricing wedges.

This paper builds on four bodies of work. The first is the literature on misallocation, which studies how wedges across heterogeneous producers distort aggregate output and welfare. Foundational contributions include Restuccia and Rogerson (2008), Hsieh and Klenow (2009), and the survey by Restuccia and Rogerson (2013). More recent work has emphasized welfare-based aggregation and clarified the distinction between technical and allocative efficiency (Petrin and Levinsohn, 2012; David and Venkateswaran, 2019; Baqaee and Farhi, 2020; Peters, 2020; Trottner, 2022; Berger et al., 2022; Bau and Matray, 2023). Particularly relevant for our purposes are Peters (2020) and Trottner (2022), who show that imperfect competition can generate misallocation through dispersion in markups across firms. We also connect to work on financial frictions and the allocation of capital across nonfinancial firms.

Midrigan and Xu (2014), Moll (2014), Gopinath et al. (2017), and Bau and Matray (2023) show that credit-market imperfections, financial integration, and the distribution of returns to capital are central for aggregate productivity. Our paper shifts the unit of analysis to banks. Banks are not only a conduit through which misallocation reaches the real economy. They are heterogeneous producers whose own market power and funding conditions shape where deposits and loans are allocated. Because banks jointly provide deposit and credit services, the relevant allocative problem is not the cross-firm allocation of a single output, but the joint allocation of deposits and loans across banks. Our framework extends the markup-misallocation logic to two-sided intermediation and decomposes welfare loss into deposit-side, loan-side, and interaction components.

The second body of work studies banking, market power, and intermediation technology. On the demand side, we build on the structural IO literature on deposit and loan demand, bank market power, and monetary transmission, including Egan et al. (2017), Drechsler et al. (2017), Wang et al. (2022), Egan et al. (2022), d’Avernas et al. (2023), Jiang et al. (2025), Koont (2025), and Whited et al. (2023). These papers quantify banks’ pricing power, funding franchise, and pass-through to borrowers and depositors. Relative to the banking IO literature, we move from pricing and pass-through to allocative efficiency and welfare. A related literature examines the movement of bank funding across markets and institutions. D’Amico and Alekseev (2025) connect interest rates to the degree of financial integration in commercial banking. Oberfield et al. (2025) and Quincy and Xu (2025) emphasize the role of branch networks and internal capital markets in shaping where deposits and loans are allocated across regions. Aguirregabiria et al. (2025) and Maingi (2025) study geographic imbalances in funding and lending and the frictions that limit the free movement of deposits across space. We study misallocation across banks rather than geographic imbalances per se. Geography matters in our setting because it shapes banks’ access to customers of different tastes, but the core object is the cross-sectional dispersion in markups across banks, even when institutions operate in the same markets. Our emphasis on markups as a source of distortion also differs from Begenau et al. (2025), who show that prudential policy can distort portfolio and scale choices across banks.

Our merger application connects to two additional literatures. The first is the macro and corporate finance literature that views firm dynamics and M&A as reallocative forces. Classic models of firm selection and the firm size distribution include Jovanovic (1982), Hopenhayn (1992), and Luttmer (2007). In corporate finance, Jovanovic and Rousseau (2002) interpret merger waves as responses to reallocation opportunities, while Maksimovic

and Phillips (2001) and Maksimovic et al. (2011) show that asset sales and post-merger restructuring tend to move assets toward owners with comparative advantage. Our merger exercise shares this reallocative view, but the object is different. We ask whether consolidated ownership moves deposit and loan pricing wedges closer to the planner’s efficient benchmark. The second is the IO literature on merger simulation, unilateral effects, and upward pricing pressure. Werden and Froeb (1994) and Nevo (2000) use demand estimates and ownership counterfactuals to simulate merger effects in differentiated-products markets. Farrell and Shapiro (2010) develop the upward pricing pressure screen, and Jaffe and Weyl (2013) provide a first-order approach that uses local information from the pre-merger equilibrium. Ex post merger studies, such as Ashenfelter and Hosken (2010) and Miller and Weinberg (2017), evaluate realized price effects and post-merger conduct. Our deal-level *ex ante* misallocation exercise borrows the ownership-counterfactual logic from this literature. The difference is that we map ownership-induced repricing into the dispersion of markups and social wedges, rather than into price increases or consumer surplus alone.

Finally, our consolidation analysis relates directly to the bank M&A literature. Berger et al. (1999) review the causes and consequences of financial consolidation. Prager and Hannan (1998) show that concentration-increasing bank mergers can lower deposit rates, while Focarelli and Panetta (2003) find that adverse deposit-rate effects can be temporary once efficiency gains materialize. On the lending side, Sapienza (2002) and Erel (2011) show that bank mergers can lower loan rates when efficiency gains dominate, but can raise them when local market power is strong. Berger et al. (1998) study small-business lending responses, and Houston et al. (2001) use market reactions and managerial projections to study the sources of bank merger gains. More recently, Granja and Paixao (2026) emphasize the role of uniform pricing in shaping the effects of bank consolidation. We complement this work by studying both *ex post* and *ex ante* effects of bank mergers on markup-driven misallocation. Our object is not whether consolidation changes average rates or market power, but whether it changes the cross-bank dispersion of deposit and loan market power that matters for allocative efficiency.

The rest of the paper is organized as follows. Section 2 presents the model and derives the welfare loss from misallocation. Section 3 describes the data and estimation. Section 4 reports the demand and welfare results. Section 5 applies the framework to bank mergers. Section 6 concludes.

## 2 The Model

We develop a model of the two-sided banking sector to formalize the notion of misallocation and to derive a set of sufficient statistics. Banks face demand on both the deposit side and the loan side. We abstract from risk, so all interest rates should be interpreted as risk adjusted.

### 2.1 Setup

#### 2.1.1 Depositors

In our baseline model, the economy contains a mass of  $M_t^D$  depositors at time  $t$ , each endowed with one dollar and choosing where to place it. The available saving options are cash, money market funds, Treasuries, and bank deposit products. Without loss of generality, we assume that each depositor chooses only one product, for example the deposit product of a single bank. Because the depositor's problem is static, we suppress the  $t$  subscript for brevity. Banks, indexed by  $j$ , offer differentiated deposit products, and each bank offers a single product. We assume a national deposit market so that all banks can potentially compete for the same pool of depositors. Under the two-sided demand system, depositor  $i$  in market  $t$  maximizes the following indirect utility from choosing bank  $j$ :

$$\max_{j \in \mathcal{A}^D} u_{ij}^D = -\alpha_i^D r_j^D + \beta^D X_j^D + \gamma_{other}^D \log s_j^L + \delta_j^D + \epsilon_{ij}^D \quad (1)$$

where  $r_j^D$  is the rate offered by bank  $j$ ,  $\alpha_i^D$  is the rate sensitivity of household type  $i$  and  $\alpha_i^D < 0$ , so depositors prefer a higher rate,  $X_j^D$  denotes the vector of bank characteristics,  $\beta^D$  is a vector of sensitivities to the non-rate characteristics,  $\gamma_{other}^D$  captures how the attractiveness of bank  $j$  on the loan side affects deposit demand,  $\delta_j^D$  is the unobservable bank-level demand shock, and  $\epsilon_{ij}^D$  is the relationship-specific shock that follows a generalized extreme-value distribution.

Importantly, we include the term  $\log s_j^L$  as a reduced-form way to model the two-sidedness of deposit-loan demand, i.e., a bank may become more attractive to depositors when it has a larger loan-side presence. This is consistent with cross-side complementarities such as one-stop banking convenience, service quality valued by both depositors and borrowers, and trust or reputation from being a major lender<sup>1</sup>. The two-sided setting has proven popular in

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<sup>1</sup>This is not a literal statement about each depositor's own borrowing decision as we use the equilibrium

media applications (Rysman, 2004; Argentesi and Filistrucchi, 2007; Fan, 2013; Song, 2021; Jullien et al., 2021).

Our bank-level two-sided framework differs from Aguirregabiria et al. (2025). Their emphasis is on the geographic flow of funds across local markets, where two-sidedness arises through local synergies between deposits and mortgage lending within a bank’s branch network and shapes the extent of home bias in lending and deposit collection. Our paper instead studies two-sidedness at the bank level. Banks jointly price deposit and loan products, and market power distorts the allocation of both sides across banks. Another empirical difference is that their loan side is limited to mortgages because of county-level data constraints, while we aim to capture banks’ broader loan portfolios.

Depositors have several outside options for placing their money, namely cash, money market funds, and Treasury bills, whose combined exponential utilities are normalized to 1. Thus, conditional on the vector of loan shares  $\mathbf{s}^L$ , the market share of bank  $j$ ’s deposits has the logit form:

$$s_j^D(r^D, \mathbf{s}^L) = \sum_{i=1}^I \nu_i^D \frac{\exp(-\alpha_i^D r_j^D + \beta^D X_j^D + \gamma_{other}^D \log s_j^L + \delta_j^D)}{1 + \sum_{n=1}^J \exp(-\alpha_i^D r_n^D + \beta^D X_n^D + \gamma_{other}^D \log s_n^L + \delta_n^D)}. \quad (2)$$

where  $\nu_i^D$  is the fraction of total wealth held by depositors of type  $i$ .

Total deposits collected by bank  $j$  are therefore

$$D_j = s_j^D M^D. \quad (3)$$

Because  $s_j^D$  depends on  $s_j^L$ , deposit and loan demand are jointly determined. In equilibrium, the vectors  $(\mathbf{s}^D, \mathbf{s}^L)$  solve the fixed-point system consisting of (2) and the loan-share equation below.

Conditional on  $\mathbf{s}^L$ , we can write the classic log-sum expression for depositors’ expected

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$\log s_j^L$  rather than an individual indicator such as “I borrow here,” but it captures the idea of two-sided banking relationships.



surplus in this class of models (McFadden, 1972; Small and Rosen, 1981; Train, 2009):

$$\begin{aligned}
CS^D &\equiv M^D \sum_{i=1}^I \nu_i^D CS_i^D \\
&= M^D \sum_{i=1}^I \nu_i^D \frac{1}{-\alpha_i^D} \log \left( 1 + \sum_{j=1}^J \exp(-\alpha_i^D r_j^D + \beta^D X_j^D + \gamma_{other}^D \log s_j^L + \delta_j^D) \right).
\end{aligned} \tag{4}$$

Holding  $\mathbf{s}^L$  fixed, this surplus satisfies:

$$\frac{\partial CS^D}{\partial r_j^D} = D_j, \tag{5}$$

which is the same as in a one-sided model, i.e., the envelope property remains valid on the deposit side, conditional on the loan-side state variable. The surplus algebra can be found in Appendix B.1.

### 2.1.2 Borrowers

We also assume a national market for loans. Under the two-sided system, borrower  $i$  has the following indirect utility from choosing bank  $j$ :

$$\max_{j \in \mathcal{A}^L} u_{ij}^L = \alpha_i^L r_j^L + \beta^L X_j^L + \gamma_{other}^L \log s_j^D + \delta_j^L + \epsilon_{ij}^L \tag{6}$$

where  $\alpha_i^L < 0$ , so borrowers dislike a higher loan rate, and  $\gamma_{other}^L$  captures how the bank's deposit-side presence affects loan demand. This can capture, for example, a preference to keep deposits where one already borrows, or expects to borrow in the future.

Borrowers can borrow from non-banks such as shadow banks and bond markets, and we again normalize the utility of such outside options to 0, so that the market share of bank  $j$ 's loans has the logit form:

$$s_j^L(r^L, \mathbf{s}^D) = \sum_{i=1}^I \nu_i^L \frac{\exp(\alpha_i^L r_j^L + \beta^L X_j^L + \gamma_{other}^L \log s_j^D + \delta_j^L)}{1 + \sum_{n=1}^J \exp(\alpha_i^L r_n^L + \beta^L X_n^L + \gamma_{other}^L \log s_n^D + \delta_n^L)}. \tag{7}$$

Loan demand for bank  $j$  is therefore

$$L_j = s_j^L M^L. \tag{8}$$

Similarly, conditional on  $\mathbf{s}^D$ , borrowers' surplus is

$$\begin{aligned} CS^L &\equiv M^L \sum_{i=1}^I \nu_i^L CS_i^L \\ &= M^L \sum_{i=1}^I \nu_i^L \frac{1}{\alpha_i^L} \log \left( 1 + \sum_{j=1}^J \exp(\alpha_i^L r_j^L + \beta^L X_j^L + \gamma_{other}^L \log s_j^D + \delta_j^L) \right), \end{aligned} \quad (9)$$

where holding  $\mathbf{s}^D$  fixed,

$$\frac{\partial CS^L}{\partial r_j^L} = -L_j. \quad (10)$$

Equations (2) and (7) together define the two-sided demand system.

### 2.1.3 Banks

Bank  $j$  finances loans  $L_j$  through deposits  $D_j$  and non-deposit liabilities  $H_j$ . The balance sheet of bank  $j$  satisfies the following condition:

$$L_j = D_j + H_j. \quad (11)$$

Taking bank characteristics  $X_j$  as given in the short run, bank  $j$  chooses the deposit rate  $r_j^D$  and loan rate  $r_j^L$  to maximize profits:

$$\Pi_j = \max_{r_j^D, r_j^L} r_j^L L_j(r^L, r^D) - r_j^D D_j(r^D, r^L) - C_j(Q_j) \quad (12)$$

where  $Q_j \equiv \{L_j, D_j\}$ , and  $C_j(Q_j)$  captures banks' operating cost of taking deposits and managing assets, as well as costs of wholesale funding or other asset-liability-management (ALM).

Different from a one-sided demand model,  $D_j$  and  $L_j$  now depend jointly on both sides' prices because of the cross-side demand terms  $\log s_j^L$  and  $\log s_j^D$ , but banks' pricing first-order

conditions remain conceptually the same:

$$[r^D] \quad r_j^D = \underbrace{-\frac{\partial C_j(Q_j)}{\partial D_j}}_{\text{net marginal value of deposits}} - \underbrace{\mu_j^D}_{\text{deposit markup}}, \quad (13)$$

$$[r^L] \quad r_j^L = \underbrace{\frac{\partial C_j(Q_j)}{\partial L_j}}_{\text{marginal cost of loans}} + \underbrace{\mu_j^L}_{\text{loan markup}}, \quad (14)$$

where the deposit markup  $\mu_j^D > 0$  measures how much bank  $j$  underpays its depositors relative to the marginal value of deposits to the bank, and the loan markup  $\mu_j^L > 0$  measures how much bank  $j$  overcharges its borrowers relative to its marginal cost.

A key distinction in our setting is that deposit markups no longer follow the standard one-sided logit formula. A change in a bank's deposit rate affects not only the quantity of deposits demanded directly but also the quantity of loans demanded through the cross-side demand linkage. That movement on the loan side then feeds back into the quantity of deposits demanded through the same channel. As a result, deposit markups depend on both the direct sensitivity of deposit demand to the deposit rate and the indirect two-sided feedback. The strength of this feedback is governed by the cross-side demand parameter  $\gamma_{other}^D$  in depositors' indirect utility in (1). The same logic applies to loans. Rewriting the FOCs gives

$$0 = \frac{\partial \Pi_j}{\partial r_j^D} = D_j - \mu_j^D \frac{\partial D_j}{\partial r_j^D} - \mu_j^L \frac{\partial L_j}{\partial r_j^D}, \quad (15)$$

$$0 = \frac{\partial \Pi_j}{\partial r_j^L} = L_j + \mu_j^D \frac{\partial D_j}{\partial r_j^L} + \mu_j^L \frac{\partial L_j}{\partial r_j^L}, \quad (16)$$

where  $\partial L_j / \partial r_j^D \neq 0$  and  $\partial D_j / \partial r_j^L \neq 0$  reflect the two-sided demand feedback. (15) and (16) further show that we can obtain deposit and loan markups using demand parameters and data on banks' rates and quantities. We collect the FOCs and related objects of all banks  $J$  into vectors. This lets us express the system compactly and derive the markups by matrix algebra:

$$\begin{bmatrix} \mu^D \\ \mu^L \end{bmatrix} = -(\Omega \odot \Delta)^{-1} \begin{bmatrix} D \\ L \end{bmatrix}, \quad (17)$$

where  $\mu^D, \mu^L, D, L$  are  $J \times 1$  stacked markups and balance sheet quantities,  $\Delta$  is the  $2J \times 2J$

matrix of quantity derivatives whose diagonal entries in each of the four blocks capture the effects of bank  $j$ 's rates on its own quantities<sup>2</sup>,  $\Omega$  is banks' ownership matrix with the  $J \times J$  identity matrix  $I_J$ , and  $\odot$  denotes the Hadamard element-wise product.

$$\Delta = \begin{bmatrix} \frac{\partial D}{\partial(-r^D)} & \frac{\partial L}{\partial(-r^D)} \\ \frac{\partial D}{\partial r^L} & \frac{\partial L}{\partial r^L} \end{bmatrix}, \quad \Omega = \begin{bmatrix} I_J & I_J \\ I_J & I_J \end{bmatrix}.$$

In the special case with no cross-side feedback, the off-diagonal blocks  $\Delta_{DL} = \Delta_{LD} = 0$ , and (17) collapses to the usual one-sided formulas separately for deposits and loans. We return to the discussion of markups with cross-side feedback in Section 4.3.

## 2.2 What is Misallocation?

We define misallocation as the welfare gap between the decentralized equilibrium and the social planner equilibrium. The planner chooses  $\{r_j^D, r_j^L\}$  for all banks to maximize total surplus ( $TS$ ):

$$\max_{r_j^D, r_j^L} TS = \sum_j^J \Pi_j + CS^D + CS^L \quad (18)$$

$$\text{s.t.} \quad \sum_j D_j = \bar{D} \quad (19)$$

$$\sum_j L_j = \bar{L} \quad (20)$$

where  $\Pi_j = r_j^L L_j - r_j^D D_j - C_j(Q_j)$ <sup>3</sup>.

A key feature of the planner's problem is that the aggregate deposits and loans are fixed at their observed levels. The planner therefore reallocates banking activity across banks rather than choosing the overall size of the banking sector. The latter would require a richer

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<sup>2</sup> $\Delta$  is slightly different from a regular quantity Jacobian  $\partial Q/\partial p$  with  $p \equiv (-r^D, r^L)$  since the off-diagonal blocks are swapped to match banks' two-sided FOCs (15) and (16). We later use  $\mathcal{J} = \partial Q/\partial p$  to avoid confusion.

<sup>3</sup>The model is sufficiently flexible to allow the planner to use a different cost function  $C_j^{SP}(Q_j)$  instead of  $C_j(Q_j)$  in the objective, so that she can strip out forces she views as distortionary in banks' private costs when re-optimizing the allocation of deposits and loans. Examples include too-big-to-fail subsidies, differential access to funding, and the shadow cost of size-based regulation, each of which would require additional and potentially much stronger modeling assumptions. We therefore set  $C_j^{SP}(Q_j) \equiv C_j(Q_j)$ , so the planner fully respects banks' private cost functions. As we discuss below, this choice changes the notion of misallocation that we measure.

model of the social value and cost of a larger banking sector and is outside the scope of this paper. Moreover, evaluating welfare differences by reallocating existing quantities keeps the exercise close to the observed data, where identification is strongest.

We first characterize the socially efficient allocation. The planner's first-order conditions (FOCs) suggest that, at the social optimum, the rates are reset so that the marginal surplus of deposits and the marginal surplus of loans are each equalized across banks<sup>4</sup>:

$$MSD_j(r_j^{D*}, Q_j^*) = \underbrace{-\frac{\partial C_j(Q_j^*)}{\partial D_j^*} - r_j^{D*}}_{\text{marginal surplus of deposits for bank } j \text{ at } Q_j^*} = \lambda^D, \quad (21)$$

$$MSL_j(r_j^{L*}, Q_j^*) = \underbrace{r_j^{L*} - \frac{\partial C_j(Q_j^*)}{\partial L_j^*}}_{\text{marginal surplus of loans for bank } j \text{ at } Q_j^*} = \lambda^L, \quad (22)$$

where  $\lambda^D$  and  $\lambda^L$  are the multipliers on the deposit  $\sum_j D_j = \bar{D}$  and loan constraints  $\sum_j L_j = \bar{L}$ , and  $r_j^{D*}$ ,  $r_j^{L*}$ , and  $Q_j^*$  are the rates and balance-sheet quantities at the social optimum.

The planner's problem differs from banks' problem by internalizing the consumer surplus terms for depositors and borrowers. However, a higher deposit rate is not valuable because it transfers resources from banks to depositors, since that transfer leaves total surplus unchanged. The marginal value of repricing comes only from the induced reallocation of deposits across banks. By raising deposit rates at some banks and lowering them at others, the planner shifts funds toward banks with a higher marginal value of deposits and thereby improves allocative efficiency by generating more social surplus with the same aggregate deposits. The same logic applies to loans.

Using banks' private FOCs with respect to  $r_j^D$  in (13), we can write (21) and (22) at the decentralized equilibrium as

$$MSD_j(r_j^D, Q_j) = -\frac{\partial C_j(Q_j)}{\partial D_j} - \left( -\frac{\partial C_j(Q_j)}{\partial D_j} - \mu_j^D \right) = \mu_j^D, \quad (23)$$

$$MSL_j(r_j^L, Q_j) = \frac{\partial C_j(Q_j)}{\partial L_j} + \mu_j^L - \frac{\partial C_j(Q_j)}{\partial L_j} = \mu_j^L, \quad (24)$$

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<sup>4</sup>Under the two-sided demand system, changing a deposit rate affects not only deposits but also loans, and vice versa. Thus, it is convenient to state the planner's first-order conditions in terms of the induced quantity allocation. Appendix B.2 shows that the conditions follow directly from the planner's first-order conditions in the original rate-setting problem.

which shows that, for bank  $j$ , the marginal surplus of deposits and loans equals the corresponding markup<sup>5</sup>.

Using deposits as an example, the planner's FOC requires all banks to have the same gap between the marginal value of deposits and the deposit rate across banks, i.e., the same deposit markup,  $\mu_j^D = \lambda^D$ . In other words, banks with a higher marginal value of deposits should offer higher deposit rates subject to a fixed pricing wedge  $\lambda^D$  and attract more deposits. When markups are not equalized, the planner raises deposit rates for banks with initially high markups and lowers them for banks with low markups. Deposits then move from socially too large banks with low markups to socially too small banks with high markups, while total deposits remain fixed.

Note that the planner does not eliminate banks' pricing power altogether through the repricing, but rather its dispersion across banks. Removing deposit market power completely would drive markups to zero rather than  $\lambda^D$ , which would reduce misallocation to zero but also change the overall size of the banking sector. This is again outside the scope of this paper. We discuss the sources of dispersion of pricing power across banks in Section 2.3.

For convenience, we define *social wedges*  $\omega^D$  and  $\omega^L$  as the gap between banks' private markups and their corresponding social optimal benchmark in a market (e.g., year)

$$\begin{aligned}\omega^D &= \mu^D - \lambda^D \mathbf{1}, \\ \omega^L &= \mu^L - \lambda^L \mathbf{1},\end{aligned}$$

where  $\mu^D$  and  $\mu^L$  are  $J \times 1$  stacked markups computed following (17),  $\lambda^D$  and  $\lambda^L$  are *scalars* of the socially optimal deposit and loan markups, and  $\mathbf{1}$  is a  $J \times 1$  vector of ones. From the planner's FOCs, we can now connect the cross-sectional dispersion of markups in a market directly to welfare loss from misallocation.

**Proposition 1** *Let  $p \equiv (-r^D, r^L)$ , and suppose  $\mathcal{J} = \partial Q / \partial p$  is locally invertible at the social optimum  $Q^*$ , and  $TS(Q)$  is twice continuously differentiable near  $Q^*$ , then the welfare gap between the social optimum and the observed decentralized equilibrium admits the local*

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<sup>5</sup>If we allow the planner to use a different cost function  $C^{SP}(\cdot)$ , then  $MSD_j(r_j^D, Q_j) = -\frac{\partial C_j^{SP}(Q_j)}{\partial D_j} - r_j^D = -\frac{\partial C_j^{SP}(Q_j)}{\partial D_j} + \frac{\partial C_j(Q_j)}{\partial D_j} + \mu_j^D = \tau_j^D + \mu_j^D$ , which indicates that the marginal surplus of deposits  $MSD_j$  for bank  $j$  can be decomposed into the deposit markup  $\mu_j^D$  and the cost wedge  $\tau_j^D$  between the net marginal value of deposits perceived by banks and that perceived by the social planner. Direct calculation gives a similar decomposition formula  $MSL_j = \tau_j^L + \mu_j^L$  for loans.

quadratic approximation

$$\begin{aligned}
TS^* - TS \approx & \underbrace{\frac{1}{2}\omega^{D\top} G_{DD}\omega^D}_{\text{dispersion of deposit markups, weighted by } G_{DD}} + \\
& \underbrace{\frac{1}{2}\omega^{L\top} G_{LL}\omega^L}_{\text{dispersion of loan markups, weighted by } G_{LL}} + \\
& \underbrace{\omega^{D\top} G_{DL}\omega^L}_{\text{interaction of deposit and loan markups, weighted by } G_{DL}}, \tag{25}
\end{aligned}$$

where the reallocation sensitivity matrix  $G \equiv \Gamma^{-1}$ , with  $\Gamma = -\nabla_Q^2 TS(Q^*)$ .

Proposition 1 follows from a second-order Taylor expansion around the socially optimal allocation of deposits  $D^*$  and loans  $L^*$ <sup>6</sup>. The first two terms capture how dispersion in social wedges within each side maps into welfare loss, while the last term captures whether deposit and loan wedges reinforce or offset each other in generating welfare loss<sup>7</sup>.

It is helpful to think of  $\omega$  as the misallocation pressure and of  $G = \Gamma^{-1}$  as the reallocation sensitivity of the banking system. Proposition 1 shows that the welfare-relevant measure of misallocation is the cross-sectional dispersion of markups, weighted by the reallocation sensitivity matrix  $G$ . The social wedge  $\omega^D$  is positive for high-markup banks that are socially too small on deposits, since their marginal surplus of deposits exceeds the planner's benchmark, and negative for banks that are socially too large. Dispersion in  $\omega^D$  therefore tells us where deposits are misallocated, but not by itself how costly that misallocation is in dollars. The latter also depends on how strongly quantities must move to close those marginal-surplus gaps, which is governed by matrix  $G$ . For instance, a lower deposit-side sensitivity matrix  $G_{DD}$  means that, for a given dispersion in  $\omega^D$ , the planner needs a smaller quantity adjustment across banks to close the gap in marginal surplus. When fewer units must be reallocated, the associated welfare loss is smaller<sup>8</sup>.

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<sup>6</sup>The formal proof can be found in Appendix B.3.

<sup>7</sup>To make the mapping clearer, we can rewrite

$$\omega^{D\top} G_{DD}\omega^D = \sum_j \sum_k \omega_j^D (G_{DD})_{jk} \omega_k^D,$$

which aggregates weighted pairwise social wedges across banks  $(j, k)$ .

<sup>8</sup>To emphasize the notion of two-sided markup-driven misallocation, our model omits the liquid securities and cash holding  $B$  of banks since they normally do not have pricing power over the securities and hence no

Economically, a lower  $G_{DD}$  can arise when demand is more inelastic, when balance-sheet costs are more convex in deposits, or more generally when the full two-sided system makes deposit adjustments less responsive to a given social wedge correction<sup>9</sup>. Formally,  $G \equiv \Gamma^{-1}$  depends on banks' demand elasticities and cost convexities because

$$\Gamma = C^{Hes} + S = \begin{bmatrix} C_{DD}^{Hes} & C_{DL}^{Hes} \\ C_{LD}^{Hes} & C_{LL}^{Hes} \end{bmatrix} + \begin{bmatrix} \frac{\partial r^D}{\partial D} & \frac{\partial r^D}{\partial L} \\ -\frac{\partial r^L}{\partial D} & -\frac{\partial r^L}{\partial L} \end{bmatrix}, \quad (26)$$

where  $C^{Hes} \equiv \nabla_Q^2 C$  denotes the Hessian matrix of the cost function with respect to deposits and loans, and  $S$  collects the inverse-demand derivatives with derivatives evaluated at  $Q^*$ , or equivalently at  $p^*$ . The off-diagonal blocks of  $\Gamma$  will be zero if deposits and loans are completely disconnected.

**Two-sidedness of Misallocation.** Banks set deposit and loan terms jointly, and the two sides of the balance sheet are linked through both demand and costs. On the demand side, relationship banking can make deposit and loan choices move together. On the cost side, funding and lending are connected through asset liability management. Two implications follow. First, deposit and loan markups are jointly determined through demand, as shown above in (15) and (16). Second, dispersion in loan markups can create a need to reallocate not only loans but also deposits, and the same logic applies in reverse. Thus, welfare loss depends not only on dispersion in deposit and loan markups taken separately, but also on the interaction between the two sides of the balance sheet.

The sign of this interaction is economically important. When  $\omega^{D\top} G_{DL} \omega^L > 0$ , deposit-side and loan-side wedges reinforce each other, so the interaction term in Proposition 1 raises total welfare loss. When  $\omega^{D\top} G_{DL} \omega^L < 0$ , the two wedges offset each other and reduce the total loss. The sign of the interaction term is determined by two forces. The first is how the signs of the two wedge vectors,  $(\omega^D, \omega^L)$ , align across banks. Reinforcement is more likely when banks that are socially too small on deposits also tend to be too small on loans, and likewise for banks that are too large on both sides. The second force is how wedges

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dispersed markups. That said, securities can still enter the welfare loss through the reallocation sensitivity  $G$  due to their role as a balance-sheet buffer. If a bank gets “too many” deposits relative to loans, it can park resources in  $B$ . If it gets “too few,” it can shrink  $B$  and rely more on wholesale funding  $H$ . Thus endogenous  $B$  changes the curvature of the planner problem.

<sup>9</sup>In a two-sided system,  $G_{DD}$  is the deposit-deposit block of the full inverse matrix,

$$G_{DD} = (\Gamma^{-1})_{DD} = (\Gamma_{DD} - \Gamma_{DL} \Gamma_{LL}^{-1} \Gamma_{LD})^{-1},$$

which depends not only on deposit-side curvature, but also on the loan-side block and the cross-side linkages.



on one side affect required quantity allocation on the other side. This is summarized by the cross-side reallocation sensitivity matrix  $G_{DL}$ , which reflects within-bank linkages from joint demand and balance-sheet costs, and also across-bank linkages under strategic Nash-Bertrand competition, since a change in one bank’s prices shifts demand across rival banks and feeds back into equilibrium reallocation. We return to the two forces in Appendix A.1 after estimating the model.

Proposition 1 opens the door to a local approach to measuring misallocation, which we prefer to a full global simulation. With our number of banks, degree of cross-sectional heterogeneity, and the Bertrand-Nash equilibrium concept, global counterfactuals can become numerically unstable. The purpose of the baseline model is to construct a measure of misallocation in banking, not to solve for an implementable transition to the social optimum. For example, the planner removes heterogeneous pricing power in the benchmark but does not model the expenses associated with doing so.

## 2.3 Economic Sources of Markup Dispersion

We focus on markup-driven misallocation, where the operative social wedge is the cross-sectional dispersion of deposit and loan markups across banks. In a differentiated-product Bertrand-Nash oligopoly, markups are pinned down by the demand elasticities and diversion patterns each bank faces. The relevant friction is therefore not market power in the aggregate, but the fact that pricing power is unevenly distributed across banks. Banks with more captive depositors or borrowers price further away from the planner’s benchmark and attract too little activity relative to the efficient allocation.

What can generate this dispersion? At a broad level, any primitive that makes bank-level demand more or less elastic can do so. One source is market segmentation. Search frictions, switching costs, relationship banking, and depositor inertia make some customers less willing to reallocate in response to small rate changes. Banks that serve these customers face less elastic demand and can sustain larger markups on deposits or larger markups on loans. A second source is heterogeneity in nonprice attractiveness. Branch networks, service quality, digital convenience, brand, trust, and local presence all shift demand toward particular banks. A bank with stronger nonprice appeal not only tends to have a larger equilibrium customer base, but may also retain relatively insensitive customers when it adjusts rates. A third source is heterogeneity in the competitive environment itself. Even with identical products, a bank that faces fewer close substitutes, operates in a more concentrated market, or competes

against weaker rivals will face softer pricing pressure. These forces are consistent with the empirical evidence that dispersion in the demographic composition and sophistication of the clientele generates heterogeneity in rate elasticities across banks (Xiao, 2020; Bhutta et al., 2026).

Some of these primitives may be interpreted as efficient and others as distortive. Better service quality, for example, may reflect real product quality, whereas switching frictions or local segmentation may reflect barriers to substitution. We do not attempt to separate these sources at the primitive level. As Proposition 1 shows, once these forces generate dispersion in markups, they create dispersion in marginal surplus across banks, and under decreasing returns that dispersion maps directly into misallocation. Thus, even when the underlying primitive is economically “efficient,” market power can make its effect on equilibrium pricing welfare-reducing.

## 3 Data and Estimation

### 3.1 Data

We use two major datasets in the empirical analysis. First, we obtain the balance sheets and income statements of banks from the quarterly Consolidated Report of Condition and Income (Call Reports), published by the Federal Financial Institutions Examination Council (FFIEC). The data on branch location and deposits are from the Summary of Deposits, published by the Federal Deposit Insurance Corporation (FDIC).

Second, we collect the county-level characteristics such as education, median income, and elderly population from the U.S. Census and County Business Patterns. The local average wages of bank tellers are from the Occupational Employment and Wage Statistics, published by the Bureau of Labor Statistics.

We keep only U.S. commercial banks engaged in traditional loan and deposit businesses from 1986 through 2021, so we drop banks with loan-to-asset ratios or deposit-to-asset ratios below 0.25 or with negative loan or deposit rates. We then winsorize the rates, shares, number of branches, and number of employees per branch at the 0.5% and 99.5% levels within each year.

## 3.2 Estimation

We estimate the model in three blocks with the goal of quantifying the welfare loss from markup-driven misallocation in banking in Proposition 1. First, we estimate the joint demand system for deposits and loans. This step identifies the demand primitives and implies the equilibrium deposit markups  $\mu^D$  and loan markups  $\mu^L$ . Second, taking the estimated demand system as given, we recover the planner’s socially optimal markups,  $\lambda^D$  and  $\lambda^L$ , from (21) and (22) using a local projection method consistent with Proposition 1. Third, we construct the reallocation sensitivity matrix  $G$  in (25), which serves as the weights that determine how dispersion in markups maps into welfare losses.

### 3.2.1 Demand Estimation

To recover the demand parameters on the deposit and loan sides, we follow the standard BLP approach to differentiated-products demand estimation (Berry et al., 1995; Nevo, 2001). Our baseline specification defines both deposit and loan markets at the national level, consistent with Wang et al. (2022) and Whited et al. (2023). This choice is primarily data-driven, since balance data for several major loan categories, such as commercial and industrial loans, are not available at finer geographic levels. We compute a bank’s deposit market share as its deposits divided by the combined deposits, cash, Treasury securities, and money market fund holdings of U.S. households and firms. We compute a bank’s loan market share as its loans divided by the total debt of U.S. households and firms. We allow demand to depend on observed bank characteristics, including the number of branches and employees per branch, and we include bank and time fixed effects to absorb persistent bank heterogeneity and common aggregate shocks.

Prices are endogenous to bank-specific demand shocks, so identification requires instruments that shift banks’ pricing incentives without directly entering local demand. Following the banking demand literature, we use cost and competitive shifters that are plausibly excluded from demand conditional on observables and fixed effects (Egan et al., 2022; Wang et al., 2022). Our instrument set draws on two sources of variation. First, we use competitors’ characteristics and cost ratios, weighted by county-overlap cosine similarity between the focal bank and its competitors. These shifters include competitors’ branch counts and their salary and fixed-asset expense ratios. The idea is that banks facing similar local competitors are exposed to similar cost and competitive conditions, which affect equilibrium prices, while these competitors’ shifters should not directly impact bank  $j$ ’s demand once we control for

bank characteristics, bank fixed effects, and time fixed effects. Second, we use predetermined balance-sheet and market-structure variables interacted with changes in the effective federal funds rate  $\Delta EFFR$ , including the lagged non-deposit liability ratio, the securities ratio (Kashyap and Stein, 2000), and deposit-market concentration index (Drechsler et al., 2017). These interactions capture heterogeneous pass-through of a common funding shock. Banks that differ *ex ante* in funding structure, asset composition, or market power should adjust prices differently in response to the same movement in EFFR.

Recall that, in our setting, the cross-side market share  $s_{jt}^{\text{other}}$  enters the demand system through  $\log(s_{jt}^{\text{other}})$ . In the deposit demand equation, this variable is the bank's loan share. In the loan demand equation, it is the bank's deposit share. We treat  $\log(s_{jt}^{\text{other}})$  as endogenous because unobserved bank-year factors may affect demand on both sides at the same time, which can make the cross-side share correlated with the structural demand shocks. We address this problem by instrumenting  $s_{jt}^{\text{other}}$  using the same excluded instruments that identify prices.

We estimate the demand system by two-step GMM<sup>10</sup>, which pins down the parameter vector that best fits the observed market shares given the prices and product characteristics in equations (2) and (7).

### 3.2.2 Estimate Socially Optimal Markups

Proposition 1 writes the welfare gap as a quadratic form in the social wedges  $\omega^D \equiv \mu^D - \lambda^D \mathbf{1}$  and  $\omega^L \equiv \mu^L - \lambda^L \mathbf{1}$ , weighted by the reallocation sensitivity matrix  $G$ . Two practical concerns are that (1) the multipliers  $\lambda^D, \lambda^L$  are not directly observed, and (2)  $G$  needs to be recovered from data and should be evaluated at the social optimum quantities  $Q^*$ . This subsection and the next show how to recover them locally, consistent with the local-quadratic approximation underlying Proposition 1. For reasons elaborated above, we prefer the local method over solving the social planner's problem globally.

We first use deposits as an example to show how we can write the planner's optimal markup  $\lambda^D$  as a  $G$ -weighted average of markups across banks. All objects are measured within a year, or market,  $t$ . To keep the notation light, we suppress the  $t$  subscript throughout this discussion. Let  $\mu^D \in \mathbb{R}^J$ ,  $\mathbf{1} \in \mathbb{R}^J$ , and let  $G \equiv G_{DD}$  denote the deposit-side welfare weight matrix in Proposition 1, where  $J$  is the number of banks in year  $t$ . For any candidate

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<sup>10</sup>To follow the best practices, we use PyBLP for estimation (Conlon and Gortmaker, 2020).

$\lambda^D \in \mathbb{R}$ , recall the social wedge

$$\omega^D \equiv \mu^D - \lambda^D \mathbf{1}.$$

Under the same local-quadratic logic used to derive Proposition 1, the first-order effect of moving deposits across banks is proportional to the weighted social wedge,

$$\Delta D \approx G \omega^D = G(\mu^D - \lambda^D \mathbf{1}). \quad (27)$$

Moreover, because the planner's problem fixes aggregate deposits, admissible reallocations must satisfy

$$\mathbf{1}^\top \Delta D = 0. \quad (28)$$

Combining (27) and (28) yields a single linear condition that pins down  $\lambda^D$ :

$$\mathbf{1}^\top G(\mu^D - \lambda^D \mathbf{1}) = 0 \implies \lambda^D = \frac{\mathbf{1}^\top G \mu^D}{\mathbf{1}^\top G \mathbf{1}}.$$

Thus  $\lambda^D$  is a  $G$ -weighted average of  $\mu^D$  across banks. This construction chooses  $\lambda^D$  so that the implied local reallocation (27) respects the aggregate deposit constraint exactly, so it is consistent with Proposition 1, which is itself a second-order (local) approximation around the planner's constrained optimum.

A key caveat so far is that, with separable deposit/loan problems, one can compute  $\lambda^D$  and  $\lambda^L$  separately using the argument above. However, because deposits and loans are linked in total surplus through two-sided demand spillovers and ALM costs, the deposit-side and loan-side reallocations must be determined jointly. The basic logic from the one-sided case above still applies. Appendix B.4 shows that  $\lambda \equiv (\lambda^D, \lambda^L)$  can be jointly computed as

$$\lambda = (A^\top G A)^{-1} A^\top G \mu, \quad (29)$$

where

$$A \equiv \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{bmatrix}, \quad \mu \equiv \begin{bmatrix} \mu^D \\ \mu^L \end{bmatrix}, \quad G \equiv \Gamma^{-1} = \begin{bmatrix} G_{DD,t} & G_{DL,t} \\ G_{LD,t} & G_{LL,t} \end{bmatrix}$$

Finally, the deposit and loan social wedges used in Proposition 1 are recovered as

$$\omega^D = \mu^D - \lambda^D \mathbf{1}, \quad \omega^L = \mu^L - \lambda^L \mathbf{1}.$$

### 3.2.3 Estimate Reallocation Sensitivity Matrix

To obtain the reallocation sensitivity matrix  $G \equiv \Gamma^{-1}$ , we first estimate the curvature matrix  $\Gamma$  by combining two objects: (1) the inverse-demand derivative  $S$  recovered from the estimated two-sided demand system, and (2) banks' cost convexity  $C^{Hes}$ . Formally, we have  $\Gamma = S + C^{Hes}$ .

**Estimate  $S$ .** The demand block is recovered locally from the estimated two-sided demand system. Starting from the market-level Jacobian of quantities with respect to prices  $p \equiv (-r^D, r^L)$ ,  $\mathcal{J} = \partial Q / \partial p$ , we invert it to obtain  $S = -\partial p / \partial Q$ . When deposit and loan demand interact through the cross-side term, the off-diagonal blocks of  $S$  are nonzero, which already captures one channel of two-sidedness.

**Estimate  $C^{Hes}$ .** Estimating the full cost function of banks requires a careful procedure (Berger and Humphrey, 1997; Berger and Mester, 1997). However, we are not primarily interested in the cost function itself but in its convexity, i.e., the second derivative. Following Wang et al. (2022) and Whited et al. (2023), we assume the cost function  $C(\cdot)$  of banks has the following parsimonious form:

$$C(D_j, L_j) = \bar{c}^D D_j + \bar{c}^L L_j + \left( f + \frac{\kappa H_j}{2 D_j} \right) H_j, \quad (30)$$

where  $H = L - D$  is the dollar quantity of wholesale funding,  $\bar{c}^D$  and  $\bar{c}^L$  denote the linear operating costs of taking deposits and making loans,  $f$  is the federal funds rate, and  $\kappa$  governs how the marginal cost of wholesale funding rises with the wholesale funding ratio  $H/D$ . This captures the idea that investors demand a higher funding premium when a bank relies more heavily on wholesale funding and is therefore perceived as riskier. The entries of  $C^{Hes}$ , namely  $C_{DD}$ ,  $C_{LL}$ , and  $C_{DL}$ , then follow directly from  $\kappa$  and the bank's balance-sheet position  $(D, L)$ , because once we take second-order derivatives, the linear terms vanish and only the curvature of the cost function remains. Our baseline analysis directly borrows the estimate of  $\kappa = 0.01$  from Wang et al. (2022) and Whited et al. (2023).

**Use  $\Gamma(Q^*)$  versus  $\Gamma(Q)$ .** Let  $Q$  denote the stacked quantity vector (for example,  $Q = (D, L)$ ). Proposition 1 is derived from a second-order expansion around  $Q^*$ , hence uses  $\Gamma(Q^*)$ , while in practice we observe  $Q$  (the decentralized equilibrium). To justify using  $\Gamma(Q)$

as a reasonable proxy for  $\Gamma(Q^*)$ , we show in the Appendix B.5 that  $\Gamma(Q)$  preserves the *second-order* accuracy of Proposition 1 up to a *third-order* remainder. The approximation accuracy is consistent with Proposition 1.

After estimating markups, optimal markups  $\lambda$ , and the reallocation sensitivity matrix  $G \equiv \Gamma^{-1}$ , we can compute the welfare loss from markup-driven misallocation and its three components in Proposition 1.

## 4 Baseline Results

### 4.1 Demand Estimates

Table 1 displays the summary statistics of our sample for demand estimation. The market shares are very small because, first, we assume a national market, and second, we keep small banks in the sample without combining them as in Wang et al. (2022).

**Table 1: Summary Statistics.**

This table reports summary statistics for the bank-year sample used in the demand estimation. Deposit and loan market shares are measured relative to the national market in each year.

| Variable                    | Mean   | SD     | P10   | P25   | P50    | P75    | P90    |
|-----------------------------|--------|--------|-------|-------|--------|--------|--------|
| Deposit rates (%)           | 2.787  | 1.799  | 0.456 | 1.171 | 2.736  | 4.025  | 5.410  |
| Loan rates (%)              | 8.238  | 2.246  | 5.274 | 6.322 | 8.315  | 9.867  | 11.230 |
| Deposit market shares (%)   | 0.007  | 0.030  | 0.000 | 0.001 | 0.002  | 0.004  | 0.010  |
| Loan market shares (%)      | 0.002  | 0.011  | 0.000 | 0.000 | 0.001  | 0.001  | 0.003  |
| No. of branches             | 7.785  | 23.121 | 1.000 | 2.000 | 3.000  | 6.000  | 12.000 |
| No. of employees per branch | 16.937 | 20.336 | 6.818 | 9.375 | 13.083 | 18.875 | 28.000 |

Table 2 shows the demand estimates on both sides. The average deposit semi-elasticity<sup>11</sup> is 0.278, which falls within the range of existing estimates, from 0.134 to 1 (Wang et al., 2022; Egan et al., 2022; Granja and Paixao, 2026). By contrast, the average loan semi-elasticity is  $-0.465$ , substantially smaller in magnitude than the estimate of  $-1.462$  in Wang et al. (2022). The gap is largely explained by differences in specification. First, we retain small banks and allow for random heterogeneity in the loan semi-elasticity  $\alpha^L$ , whereas Wang et al. (2022) aggregate small banks and assume a homogeneous elasticity of loan rates across borrowers and banks. Second, and likely more important, our loan demand specification

<sup>11</sup>The average semi-elasticity  $\mathbb{E}[\alpha_i^D]$  for deposits is negative because we define the deposit price as  $-r^D$ .

includes the instrumented deposit share  $s^D$ , introducing cross-side demand dependence that is not incorporated in their setup<sup>12</sup>.

The own-price estimates also suggest a clear asymmetry across the two sides of the balance sheet. On average, borrowers are more price sensitive than depositors. At the same time, depositor price sensitivity is much more heterogeneous, with a standard deviation of 0.447 compared with 0.220 on the loan side. This pattern is intuitive in a national market where loan demand responds more sharply to rate differences, while deposit demand is relatively sticky on average but varies substantially across customer types. The large dispersion on the deposit side is also consistent with evidence that dispersion in depositor demographics and financial sophistication generate meaningful heterogeneity in rate sensitivity across banks (Xiao, 2020; Bhutta et al., 2026). Nonprice characteristics matter on both sides as well. More branches and more employees per branch raise demand for both deposits and loans, with somewhat larger effects on the deposit side. This suggests that branch networks and service capacity remain central to the deposit franchise even when banks compete in a national market.

A particularly important result is the positive and significant  $\gamma_{other}$  term in both equations (0.655 in the loan equation and 0.549 in the deposit equation). Conditional on own price, observed bank characteristics, and bank and time fixed effects, a bank with a larger presence on the other side of the balance sheet attracts more demand on the focal side. The estimates are consistent with cross-side complementarities in banking, where depositors value banks with a stronger lending franchise, and borrowers value banks with a stronger deposit franchise. Economically, this can capture one-stop banking, relationship capital, trust, and cross-selling. The effect is stronger in the loan equation than in the deposit equation, suggesting that a strong deposit franchise is especially useful for attracting borrowers.

## 4.2 Welfare Estimates

We now discuss the local welfare implications of the markup-driven misallocation summarized by Proposition 1. We focus on banks with more than 10 branches to improve numerical stability and to reduce the computational burden. This restriction is especially helpful for high-dimensional operations such as matrix inversion, which otherwise slow the analysis. This sample includes 653 to 929 banks, which account for 66% to 87% of total bank assets

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<sup>12</sup>We find more similar estimates to Wang et al. (2022) when we collapse the small banks into a hypothetical fringe bank or remove the cross-side shares.



**Table 2: Demand Estimates**

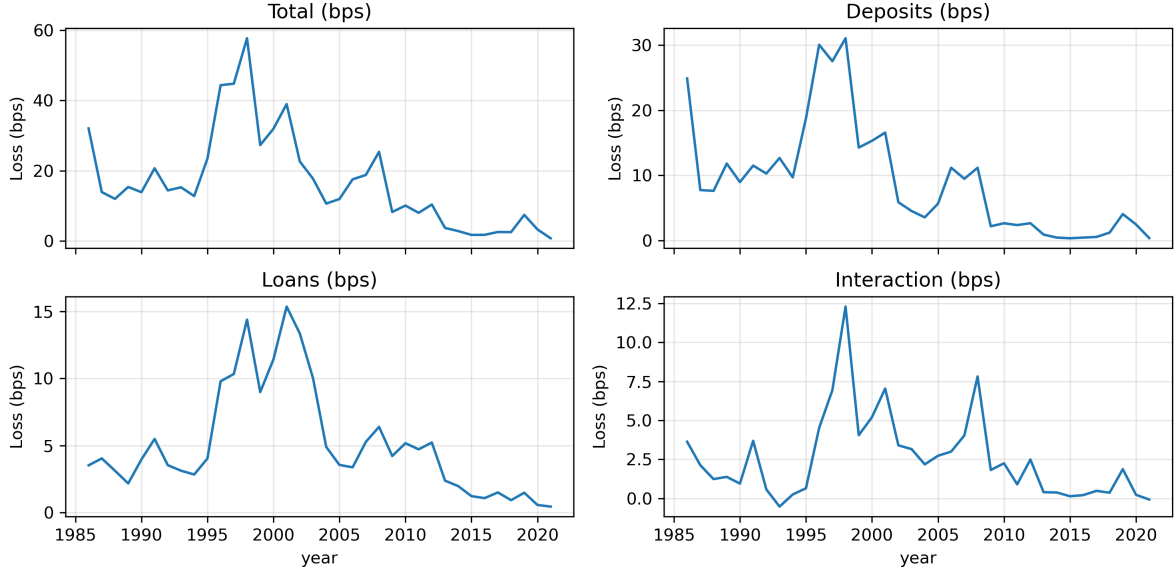
This table reports the BLP estimates of the two-sided demand system for deposits and loans. The deposit and loan equations are estimated at the national-year level, allowing demand on each side to depend on the same bank’s share on the other side. The average semi-elasticity  $\mathbb{E}[\alpha_i^D]$  for deposits is negative because we define the deposit price as  $-r^D$ . Standard errors for the estimated parameters are clustered at the bank level and reported in brackets.

| Parameter              | Description                         | Estimate | S.E.    |
|------------------------|-------------------------------------|----------|---------|
| Loan Demand            |                                     |          |         |
| $\mathbb{E}[\alpha_i]$ | Avg. sensitivity to price           | -0.465   | (0.233) |
| $\text{SD}(\alpha_i)$  | Std. dev. of sensitivity to price   | 0.220    | (0.126) |
| $\beta_{\log(nbr)}$    | Sensitivity to number of branches   | 0.380    | (0.036) |
| $\beta_{\log(empbr)}$  | Sensitivity to employees per branch | 0.384    | (0.034) |
| $\gamma_{other}$       | Sensitivity to cross-side share     | 0.655    | (0.041) |
| Deposit Demand         |                                     |          |         |
| $\mathbb{E}[\alpha_i]$ | Avg. sensitivity to price           | -0.278   | (0.044) |
| $\text{SD}(\alpha_i)$  | Std. dev. of sensitivity to price   | 0.447    | (0.153) |
| $\beta_{\log(nbr)}$    | Sensitivity to number of branches   | 0.333    | (0.062) |
| $\beta_{\log(empbr)}$  | Sensitivity to employees per branch | 0.289    | (0.059) |
| $\gamma_{other}$       | Sensitivity to cross-side share     | 0.549    | (0.067) |

in each year. The remaining *fringe* banks affect the welfare calculations only indirectly through the demand system, and their own entries in the social wedges  $\omega$  and reallocation sensitivity matrix  $G$  are excluded from the analysis below. Economically, this means that deposits and loans can be reallocated only among the selected banks, while the fringe banks remain fixed. Estimated markups are winsorized within each year at the 2.5% levels. For ease of interpretation, we scale the total welfare loss and its components by total assets. All estimates are reported in basis points.

Three main results emerge from Figures 1 and 2. First, the local welfare cost of markup-driven misallocation is economically meaningful early in the sample but declines sharply over time. The total annual loss peaks in the late 1990s at about 0.6 percent of total assets (or \$22.3 billion in 2019 dollars), then falls steadily and approaches zero in the most recent years. One possible explanation for the peak is the wave of bank mergers in the late 1990s, which we study more directly in Section 5. The muted losses in recent years may instead reflect the low-rate environment and the effective zero lower bound, which compresses pricing distortions.

Second, deposit-side distortions account for most of the welfare loss in most years, which



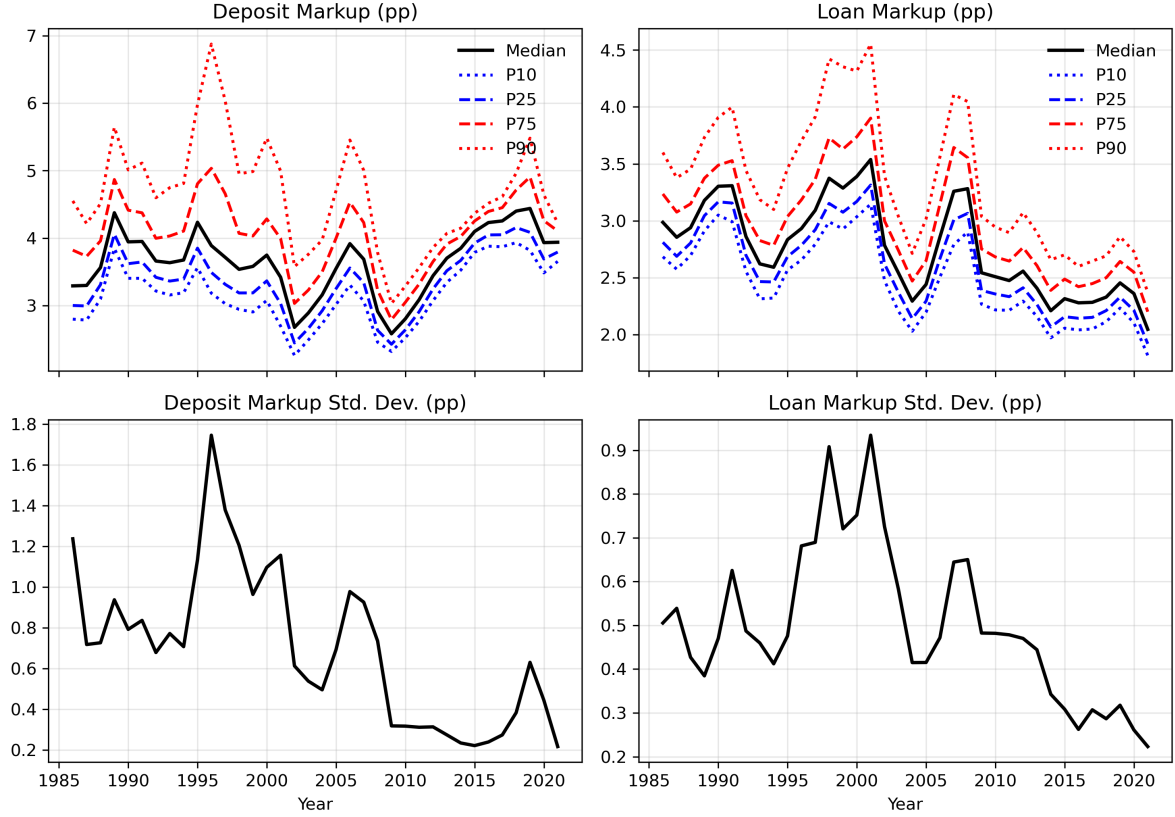
**Figure 1: Welfare Loss over Time**

This figure plots the estimated welfare loss from markup-driven misallocation over time. The series decompose the total loss into deposit-side, loan-side, and interaction components as in Proposition 1. Estimated markups are winsorized within each year at the 2.5% level. We scale the total welfare loss and its components by total assets. All estimates are reported in basis points.

suggests that inefficient deposit allocation has been the main source of banking misallocation in our estimates. Figure 2 points to the same conclusion. The cross-sectional dispersion of deposit markups is generally larger than that of loan markups, and its time-series movement closely mirrors the evolution of total welfare loss<sup>13</sup>. This pattern is driven mainly by the greater heterogeneity in deposit rate sensitivity across depositors, captured by  $SD(\alpha_i^D)$ . In other words, depositors exhibit more dispersed tastes over banks than borrowers do. Overall, the decline in misallocation over time largely reflects a compression in the cross-sectional dispersion of markups on both the deposit and loan sides.

Third, the interaction component is positive in most years, and especially pronounced in the late 1990s and mid-2000s, implying that deposit and loan wedges generally amplify rather than attenuate the welfare costs of misallocation. Its overall magnitude remains moderate relative to the total loss but still reaches about \$4.8 billion (in 2019 dollars) per year at its peak. We further examine what drives the sign of the interaction term over time in Appendix

<sup>13</sup>Average markups are relatively stable over time and exhibit no clear trend, in contrast to the decreasing raw rates reported in Figure 7 in Appendix A. The raw loan rates are also more dispersed than deposit rates, which is the opposite of what we find for markups. These differences make clear that recovering markups from demand estimation is not mechanical.



**Figure 2: Markups over Time**

This figure plots the cross-sectional distribution of deposit and loan markups over time. Deposit and loan markups are computed jointly as banks internalize the two-sided demand complementarities. Estimated markups are winsorized within each year at the 2.5% level. All estimates are reported in percentage points.

**A.1.** One channel is the alignment of bank-level social wedges across the two sides of the balance sheet. If the same banks are socially too small or too large in both deposits and loans, then deposit and loan wedges tend to have the same sign. The other channel is the cross-side reallocation sensitivity  $G^{DL}$ . This term is shaped by own-bank and cross-bank demand elasticities together with the curvature of banks' balance-sheet costs, and it governs how a distortion on one side is transmitted into reallocation on the other side. We find that the main source of time variation is the changing alignment of deposit and loan wedges across banks. By contrast, the transmission mechanism captured by  $G^{DL}$  is much more stable over time.

Overall, the time series evidence shows that, as expected, the cross-sectional dispersion of deposit and loan markups, together with their correlation, is highly informative about both the level and the composition of welfare losses from markup-driven misallocation over

time.

### 4.3 Counterfactuals

We use a targeted set of counterfactual experiments to isolate how the model primitives shape both the dispersion of markups across banks and the welfare consequences of such dispersion. Table 3 reports the average welfare loss and the standard deviation of markups from 1986 to 2006 under two classes of scenarios. We first shut down the cross-side demand link and keep all other estimated parameters fixed, which isolates the role of two-sided demand in creating markup dispersion and cross-side transmission. We then vary  $\kappa$  to isolate the role of wholesale cost convexity, the other force that connects loans and deposits, in mapping a given dispersion in markups to welfare costs.

**Table 3: Counterfactual Experiments**

This table reports the results of a set of counterfactual experiments in which we examine the impacts of different parameter values on the welfare loss from markup-driven misallocation. The first column lists the scenarios. The next four columns present the total welfare loss normalized by total assets, in basis points, and its three components, all averaged during the period 1986–2006. The last two columns present the standard deviation of deposit and loan markups in percentage points.  $\kappa = 0.01$  in the baseline and the other default parameter values are reported in Table 2.

| Scenario                                  | Total<br>(bps) | Deposits<br>(bps) | Loans<br>(bps) | Interaction<br>(bps) | Deposit<br>Markup<br>Std. Dev.<br>(pp) | Loan<br>Markup<br>Std. Dev.<br>(pp) |
|-------------------------------------------|----------------|-------------------|----------------|----------------------|----------------------------------------|-------------------------------------|
| Baseline                                  | 24.073         | 14.016            | 6.733          | 3.324                | 0.923                                  | 0.576                               |
| $\gamma_{other}^D = \gamma_{other}^L = 0$ | 32.935         | 20.125            | 10.969         | 1.841                | 1.420                                  | 0.927                               |
| $\kappa = 0$                              | 26.575         | 16.748            | 7.002          | 2.825                | 0.923                                  | 0.576                               |
| $\kappa = 0.02$                           | 22.701         | 12.676            | 6.619          | 3.406                | 0.923                                  | 0.576                               |

The table shows that shutting down the cross-side demand link by setting  $\gamma_{other}^D = \gamma_{other}^L = 0$  raises the total welfare loss from 24.1 to 32.9 basis points of assets. The increase comes from both the deposit and loan components. The deposit component rises from 14.0 to 20.1 basis points, and the loan component rises from 6.7 to 11.0 basis points. By contrast, the interaction component falls from 3.3 to 1.8 basis points. Thus, the net effect of the estimated cross-side linkages is to lower total misallocation, even though they raise the contribution of the interaction margin. This is consistent with the pattern in markup dispersion. When  $\gamma_{other}^D$  and  $\gamma_{other}^L$  are set to zero, the standard deviation of deposit markups rises from 0.923

to 1.420 percentage points, while the standard deviation of loan markups rises from 0.576 to 0.927 percentage points.

Thus, in our estimates, cross-side demand links compress heterogeneity in pricing power within each side. A positive cross-side coefficient  $\gamma_{other}$  means that raising the markup on one side becomes more costly because it reduces demand on the other side as well. This force lowers markups for all banks, but it reduces dispersion only when the induced penalty is larger for banks that would otherwise choose relatively high markups, and vice versa. In that case, cross-side feedback acts more strongly on the upper tail and pulls markups closer together. This is not mechanical. If the effective cross-side benefit of expanding the other side is much larger for some banks than for others, the same linkage can widen rather than compress markup dispersion. The cross-side benefit depends on bank-level shares, elasticities, substitution patterns, and cost curvature.

At the same time, the nonzero  $\gamma_{other}^D$  and  $\gamma_{other}^L$  strengthen the cross-side transmission of distortions in welfare, which raises the interaction component of the welfare loss. In principle, this could happen for two reasons. Deposit and loan wedges could become more aligned across banks, or the off-diagonal reallocation sensitivity block  $G_{DL}$  could become larger in magnitude. Our estimates support the second channel rather than the first. An untabulated analysis shows that the baseline displays a lower cross-sectional correlation between  $\omega^D$  and  $\omega^L$ , as well as a lower share of bank-year observations in which the two wedges have the same sign, than the no-feedback ( $\gamma_{other}^D = \gamma_{other}^L = 0$ ) case. Thus, the larger interaction term in the baseline does not arise because connected demand makes banks more likely to have high or low markup wedges on both sides at the same time. It arises because, once cross-side demand links are present, a markup distortion on one side generates a larger required quantity reallocation on the other side.

Cross-side demand therefore has two effects. It compresses markup dispersion within each side, which lowers the deposit and loan components of welfare loss. But it also increases the magnitude of cross-side propagation, which raises the interaction component. In our data, the first force dominates the second, so the estimated cross-side demand links reduce overall welfare loss. Taken together, the counterfactuals show that cross-side demand links matter not only for the level of markups, but also for their cross-sectional dispersion within each side, which is the margin most relevant for misallocation and has received much less attention in the literature.

On the other hand, the parameter  $\kappa$  links loans and deposits through the cost function. Specifically,  $\kappa$  governs the convexity of balance-sheet costs and hence the reallocation sen-

sitivity  $G$  of the system. When  $\kappa = 0$ , when banks' cost function is completely linear in deposits and loans and two sides are disconnected through cost, the average total loss rises modestly from 24.1 to 26.6 basis points. When  $\kappa = 0.02$ , the corresponding losses fall to 22.7 basis points. The deposit markup dispersion is unchanged across these cases, which is exactly what the model predicts, since  $\kappa$  does not alter the measured markups in the demand system but changes only how costly a given set of wedges is.

This welfare pattern again connects to our interpretation of Proposition 1. A larger  $\kappa$  makes costs more convex, so smaller quantity movements are sufficient for the planner to narrow a given set of markup gaps. In other words, a higher cost convexity lowers reallocation sensitivity and reduces the welfare loss attached to the same observed dispersion in markups, and a smaller  $\kappa$  has the opposite effect. Quantitatively, however, these changes are relatively modest<sup>14</sup>.

## 5 Misallocation and Consolidation

Historically, U.S. banking markets have undergone major waves of consolidation. We use our framework to study whether bank mergers increase markup-driven misallocation *ex post*. This question is distinct from the usual focus of the M&A literature, which asks whether mergers raise concentration, market power, and the level of markups. We focus on whether realized mergers increase the dispersion of markups.

We use a stacked event-study design to answer this question. The main outcome variables are the absolute social wedges of bank  $j$  at year  $t$ ,

$$|\hat{\omega}_{jt}^{D,-g}|, \quad |\hat{\omega}_{jt}^{L,-g}|, \quad |\hat{\omega}_{jt}^{D,-g}| + |\hat{\omega}_{jt}^{L,-g}|,$$

where the superscript  $-g$  indicates that, for banks involved in merger  $g$ , we compute the benchmark using a leave-merger-out analogue of  $\lambda_t$  and then obtain the leave-merger-out social wedges through the same markup demeaning  $\omega_{jt}^{D,-g} = \mu_{jt}^D - \lambda_t^{D,-g}\mathbf{1}$  and  $\omega_{jt}^{L,-g} = \mu_{jt}^L - \lambda_t^{L,-g}\mathbf{1}$ . This avoids mechanically moving the benchmark toward the treated banks after the deal. Although the absolute social wedges are *not* welfare measures because they ignore the welfare weight  $G_t$ , they are still informative. As we show in Section 4.2, the time

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<sup>14</sup>This is reassuring for two reasons. First, our main findings are not especially sensitive to the exact value of  $\kappa$ , which we borrow from the literature. Second, the comparison helps separate two roles in the model. The demand system, especially on the deposit side, is what primarily determines the welfare cost of markup-driven misallocation.

series of markup dispersion tracks that of welfare loss fairly closely.

We complement the absolute social wedges with the bank-level welfare loss contribution, constructed as the exact row-wise decomposition of the local quadratic welfare loss in Proposition 1. For a given year  $t$ , let the leave-merger-out welfare loss  $L_t^{-g}$  be

$$L_t^{-g} = \frac{1}{2}(\omega_t^{D,-g})^\top G_{DD,t} \omega_t^{D,-g} + \frac{1}{2}(\omega_t^{L,-g})^\top G_{LL,t} \omega_t^{L,-g} + (\omega_t^{D,-g})^\top G_{DL,t} \omega_t^{L,-g}.$$

where we use the same leave-merger-out social wedges  $\omega_t^{-g} \in \mathbb{R}^J$  defined above, and the original reallocation sensitivity matrix  $G$  as the welfare weight. We then rewrite  $L_t^{-g}$  and assign this quadratic form to banks row by row,

$$L_t^{-g} = \sum_j \ell_{jt}^{D,-g} + \sum_j \ell_{jt}^{L,-g} + \sum_j \ell_{jt}^{X,-g}$$

where the deposit-side, loan-side, and the interaction contributions for bank  $j$  in year  $t$  are

$$\begin{aligned} \ell_{jt}^{D,-g} &= \frac{1}{2} \omega_{jt}^{D,-g} (G_{DD,t} \omega_t^{D,-g})_j, & \ell_{jt}^{L,-g} &= \frac{1}{2} \omega_{jt}^{L,-g} (G_{LL,t} \omega_t^{L,-g})_j, \\ \ell_{jt}^{X,-g} &= \frac{1}{2} \omega_{jt}^{D,-g} (G_{DL,t} \omega_t^{L,-g})_j + \frac{1}{2} \omega_{jt}^{L,-g} (G_{LD,t} \omega_t^{D,-g})_j. \end{aligned}$$

The total bank-level welfare contribution is then

$$\ell_{jt}^{-g} = \ell_{jt}^{D,-g} + \ell_{jt}^{L,-g} + \ell_{jt}^{X,-g},$$

where, intuitively,  $\ell_{jt}^{-g}$  measures how much bank  $j$  contributes to the welfare loss once its own deposit and loan wedges are weighted by the equilibrium reallocation sensitivities embedded in  $G_t$ .

We use a stacked event-study design, which is well suited to staggered treatment timing and helps avoid contamination from already-treated units (Cengiz et al., 2019; Miller, 2023; Baker et al., 2022). We define merger cohort  $c$  by deal year  $T_c$  and study each cohort over the event window  $[-3, 2]$  around  $T_c$ . For each cohort  $c$ , we then construct a bank-year panel that includes the acquirers and a control group of banks that are not involved in any merger within the expanded window  $[-7, 7]$  around  $T_c$ . If a bank  $j$  acquires multiple targets in a year, we collapse them down to one observation per acquirer-cohort  $jc$ . We then estimate

$$|\widehat{\omega}_{jct}^{-g}| = \beta \text{ACQUIRER}_{jc} \times \text{POST}_{ct} + \Phi' \left( X_{jc}^{\text{pre}} \times \text{POST}_{ct} \right) + \alpha_{jc} + \delta_{ct} + \varepsilon_{jct}, \quad (31)$$

where  $\text{ACQUIRER}_{jc} = 1$  if bank  $j$  is an acquirer in cohort  $c$ , and equals 0 if bank  $j$  belongs to the control group in cohort  $c$ ,  $\text{POST}_{ct} = 1$  in the deal year and in all subsequent years,  $\alpha_{jc}$  are cohort-bank fixed effects, and  $\delta_{ct}$  are cohort-calendar-year fixed effects. The control variables  $X_{jc}^{\text{pre}}$  include log assets, equity ratio, and the standard deviation of quarterly ROA in the past 2 years. Each is fixed at its pre-merger level at  $T_c - 3$  for each bank and then interacted with  $\text{POST}_{ct}$ , so we allow wedge trends to differ by pre-existing bank characteristics.  $\beta$  is the average post-merger “treatment” effect. A positive  $\beta$  means that after the merger, the acquirers’ markups move further from the leave-merger-out benchmark than the control group’s wedges do in the same event window.

Because the dependent variables are generated from the estimated demand system and the welfare decomposition, the reported clustered standard errors should be interpreted as conditional on the first-stage estimates. This may understate total sampling uncertainty. In ongoing work, we are implementing a first-stage propagation bootstrap that redraws demand parameters and reconstructs the merger outcomes.

**Table 4: Bank Mergers and Absolute Social Wedges**

This table reports estimates from the stacked regressions that relate banks’ M&A activity to the absolute gap between markups and the social benchmark.  $\text{ACQUIRER}_{jc} = 1$  if bank  $j$  is an acquirer in cohort  $c$ , and equals 0 if bank  $j$  belongs to the control group in cohort  $c$ .  $\text{POST}_{ct} = 1$  in the deal year and in all subsequent years. The control variables include log assets, equity ratio, and the standard deviation of quarterly ROA in the past 2 years. Each is fixed at its pre-merger level at  $T_c - 3$  for each bank and then interacted with  $\text{POST}_{ct}$ . The standard errors in parentheses are clustered by bank and cohort. \*, \*\*, and \*\*\* denote significance at the 10%, 5%, and 1% level, respectively.

| Dependent Variables:                 | $ \hat{\omega}_{jt}^{D,-g} $<br>(1) | $ \hat{\omega}_{jt}^{L,-g} $<br>(2) | $ \hat{\omega}_{jt}^{D,-g}  +  \hat{\omega}_{jt}^{L,-g} $<br>(3) |
|--------------------------------------|-------------------------------------|-------------------------------------|------------------------------------------------------------------|
| $\text{ACQUIRER} \times \text{POST}$ | 0.078* (0.043)                      | 0.113*** (0.034)                    | 0.198*** (0.066)                                                 |
| Controls                             | Yes                                 | Yes                                 | Yes                                                              |
| Cohort-Bank FE                       | Yes                                 | Yes                                 | Yes                                                              |
| Cohort-Year FE                       | Yes                                 | Yes                                 | Yes                                                              |
| Observations                         | 27,948                              | 27,948                              | 27,948                                                           |
| Adjusted $R^2$                       | 0.905                               | 0.823                               | 0.873                                                            |
| Average Y                            | 1.926                               | 0.852                               | 2.790                                                            |

Following the same sample restrictions as in the welfare analysis, we keep only acquirers with more than 10 branches so that social wedges are well defined. We also exclude acquirers whose targets all have 10 or fewer branches, because in later analysis we use the target’s social wedge and want to maintain a consistent sample. Thus, our final sample focuses on larger mergers, includes 338 unique acquirers and 560 unique control banks, and covers



eligible merger deals from 1988 to 2019. The social wedges and welfare contribution are winsorized within each cohort and year at the 2.5% level.

Table 4 shows that, on average, mergers increase the absolute social wedge of acquiring banks relative to control banks, i.e., acquirers' markups move further from the socially efficient benchmark after the merger. The average treatment effect is around 7.1% of the mean absolute two-sided wedges. We find consistent results in Table 5, where the outcome variables are the bank-year welfare loss contribution normalized by total assets. On the deposit side, the loan side, and the interaction component, the average effects of mergers on the acquirers' contribution to welfare loss are economically and statistically significant.

**Table 5: Bank Mergers and Contribution to Welfare Loss**

This table reports estimates from the stacked regressions that relate banks' M&A activity to their contribution to welfare loss in Proposition 1.  $\text{ACQUIRER}_{jc} = 1$  if bank  $j$  is an acquirer in cohort  $c$ , and equals 0 if bank  $j$  belongs to the control group in cohort  $c$ .  $\text{POST}_{ct} = 1$  in the deal year and in all subsequent years. The control variables include log assets, equity ratio, and the standard deviation of quarterly ROA in the past 2 years. Each is fixed at its pre-merger level at  $T_c - 3$  for each bank and then interacted with  $\text{POST}_{ct}$ . The standard errors in parentheses are clustered by bank and cohort. \*, \*\*, and \*\*\* denote significance at the 10%, 5%, and 1% level, respectively.

| Dependent Variables:    | $\widehat{\ell}_{jt}^{D,-g}$<br>(1) | $\widehat{\ell}_{jt}^{L,-g}$<br>(2) | $\widehat{\ell}_{jt}^{X,-g}$<br>(3) | $\widehat{\ell}_{jt}^{Total,-g}$<br>(4) |
|-------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-----------------------------------------|
| ACQUIRER $\times$ POST  | 0.153*** (0.047)                    | 0.072*** (0.023)                    | 0.049** (0.018)                     | 0.298*** (0.084)                        |
| Controls                | Yes                                 | Yes                                 | Yes                                 | Yes                                     |
| Cohort-Bank FE          | Yes                                 | Yes                                 | Yes                                 | Yes                                     |
| Cohort-Year FE          | Yes                                 | Yes                                 | Yes                                 | Yes                                     |
| Observations            | 27,948                              | 27,948                              | 27,948                              | 27,948                                  |
| Adjusted R <sup>2</sup> | 0.595                               | 0.398                               | 0.546                               | 0.570                                   |
| Average Y               | 0.108                               | 0.033                               | 0.033                               | 0.179                                   |

We further estimate the dynamic effects of bank mergers

$$\begin{aligned}
|\widehat{\omega}_{jct}^{-g}| = & \sum_{h \neq -1} \beta_h \mathbf{1}\{t - T_c = h\} \times \text{ACQUIRER}_{jc} \\
& + \sum_{h \neq -1} \phi_h \mathbf{1}\{t - T_c = h\} \times X_{jc}^{pre} + \alpha_{jc} + \delta_{ct} + \varepsilon_{jct},
\end{aligned} \tag{32}$$

where the omitted event time is  $h = -1$ . Under parallel trends, the post-merger coefficients  $\beta_h$  trace the average effect of becoming an acquirer on the absolute social wedges. A negative average post-merger effect implies that M&A compresses wedge dispersion on average, and a positive effect implies the opposite. To address potential contamination in the dynamic

treatment effect estimates from staggered treatment timing and heterogeneous treatment effects, we use the robust estimator developed by [Sun and Abraham \(2021\)](#).

Table 6 presents the results. Relative to  $h = -1$ , acquirers' absolute social wedges and welfare contributions rise after the merger more than those of the control banks, which is consistent with the pooled results. At the same time, the event-study coefficients show a clear pre-trend. Before the merger, acquirers have significantly lower absolute social wedges and smaller welfare contributions than the control banks. One natural interpretation is selection. Banks that become acquirers are not random. They are typically larger, more productive, and closer to the planner benchmark before the deal. Therefore, the post-merger increase is best read as a deterioration in allocative efficiency relative to the acquirer's pre-merger position, not as evidence that acquirers start from the same level as the controls.

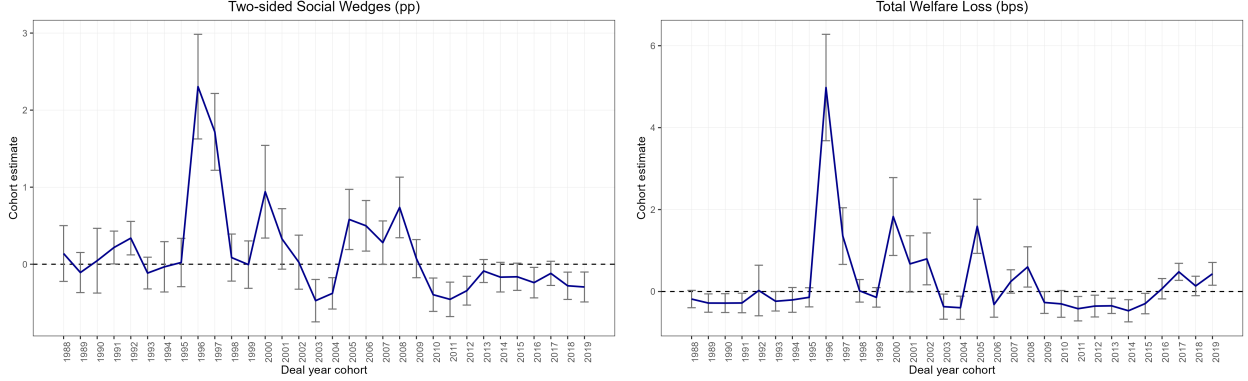
**Table 6: Bank Mergers and Absolute Social Wedges – A Dynamic Model**

This table reports the [Sun and Abraham \(2021\)](#) robust estimates from the dynamic stacked regressions that relate banks' M&A activity to the absolute gap between markups and the social benchmark.  $ACQUIRER_{jc} = 1$  if bank  $j$  is an acquirer in cohort  $c$ , and equals 0 if bank  $j$  belongs to the control group in cohort  $c$ . Event Year =  $h$  equals 1 when it is  $h$  year after the merger. The control variables include log assets, equity ratio, and the standard deviation of quarterly ROA in the past 2 years. Each is fixed at its pre-merger level at  $T_c - 3$  for each bank and then interacted with  $POST_{ct}$ . The standard errors in parentheses are clustered by bank and cohort. \*, \*\*, and \*\*\* denote significance at the 10%, 5%, and 1% level, respectively.

| Dependent Variables:                   | $ \hat{\omega}_{jt}^{D,-g}  +  \hat{\omega}_{jt}^{L,-g} $<br>(1) | $\hat{\ell}_{jt}^{Total,-g}$<br>(2) |
|----------------------------------------|------------------------------------------------------------------|-------------------------------------|
| ACQUIRER $\times$ Event Year = -3      | -0.094*** (0.027)                                                | -0.012* (0.007)                     |
| ACQUIRER $\times$ Event Year = -2      | -0.055** (0.021)                                                 | -0.019*** (0.005)                   |
| ACQUIRER $\times$ Event Year = 0       | -0.003 (0.085)                                                   | -0.106 (0.118)                      |
| ACQUIRER $\times$ Event Year = 1       | 0.284*** (0.088)                                                 | 0.454*** (0.132)                    |
| ACQUIRER $\times$ Event Year = 2       | 0.387*** (0.093)                                                 | 0.764*** (0.155)                    |
| Controls                               | Yes                                                              | Yes                                 |
| Cohort-Bank FE                         | Yes                                                              | Yes                                 |
| Cohort-ACQUIRER $\times$ Event Year FE | Yes                                                              | Yes                                 |
| Observations                           | 27,948                                                           | 27,948                              |
| Adjusted R <sup>2</sup>                | 0.880                                                            | 0.618                               |
| Average Y                              | 2.790                                                            | 0.179                               |

We next estimate cohort-specific average treatment effects using the event-study estimator of [Sun and Abraham \(2021\)](#), where cohorts are defined by merger deal year. This specification allows treatment effects to vary flexibly across cohorts and reveals substantial heterogeneity over time, as shown in Figure 3. Mergers in the 1990s tend to worsen misallocation, as these cohorts exhibit post-merger increases in both the two-sided social wedge and

total welfare loss, with several cohorts showing economically large effects. These estimates are consistent with the rise in misallocation-related welfare loss during the 1990s shown in Figure 1. By contrast, the effects are much smaller for later cohorts and are broadly neutral, or even negative.



**Figure 3: Cohort-Specific Effects of Bank Mergers on Social Wedges and Welfare Loss**

This figure plots the cohort-specific merger effects estimated from the [Sun and Abraham \(2021\)](#) event-study specification. The left panel shows the estimated effect on the sum of the two-sided social wedge measure, while the right panel shows the estimated effect on total welfare loss in basis points of total assets. Each point corresponds to a deal year cohort, and the vertical bars report 95 percent confidence intervals.

Finally, to highlight heterogeneity across deals, we construct an *ex ante* measure of the predicted change in misallocation loss  $\widehat{\Delta L}_g$  for each merger deal  $g$ , using only data from the pre-merger window  $[-3, -1]$ . The measure is based on a simple ownership counterfactual where we keep fixed the pre-merger demand system and observed quantities, and change only the ownership structure so that the acquirer and target jointly internalize the profits of both banks. Intuitively, once the two banks are under common ownership, deposits or loans diverted from one to the other are no longer treated as a loss to a competitor. This changes pricing incentives and, in turn, the markups implied by the model. We then translate the resulting change in markup dispersion across banks into the deal-level predicted change in welfare loss from misallocation. Appendix [A.2](#) describes the construction in detail. A positive  $\widehat{\Delta L}_g$  means the deal is predicted to raise the welfare loss from markup-driven misallocation, while a negative  $\widehat{\Delta L}_g$  means the deal is predicted to reduce it. We then sum the deal-level predicted welfare losses to the acquirer-deal year level and test whether acquirers with larger predicted welfare losses *ex ante* show larger extitex post deviations of markups from the social benchmark. Tentative results in Table 7 confirm this prediction<sup>15</sup>.

<sup>15</sup>The results are tentative as estimated regressors can bias the coefficient through classic or non-classic

**Table 7: Bank Mergers and Absolute Social Wedges – Heterogeneous Effects**

This table reports estimates from the stacked regressions that relate banks’ M&A activity to the absolute gap between markups and the social benchmark.  $ACQUIRER_{jc} = 1$  if bank  $j$  is an acquirer in cohort  $c$ , and equals 0 if bank  $j$  belongs to the control group in cohort  $c$ .  $POST_{ct} = 1$  in the deal year and in all subsequent years. The control variables include log assets, equity ratio, and the standard deviation of quarterly ROA in the past 2 years. Each is fixed at its pre-merger level at  $T_c - 3$  for each bank and then interacted with  $POST_{ct}$ .  $\Delta^{bps} L_g^{Total}$  is the deal-level change in welfare loss in bps, implied by ownership-induced repricing holding fixed the pre-merger allocation, demand system, and welfare weights.  $\Delta^{bps} L^{Total} < 0$  indicates a reduction in markup-driven misallocation. The standard errors in parentheses are clustered by bank and cohort. \*, \*\*, and \*\*\* denote significance at the 10%, 5%, and 1% level, respectively.

| Dependent Variables:                                       | $ \hat{\omega}_{jt}^{D,-g}  +  \hat{\omega}_{jt}^{L,-g} $<br>(1) | $\hat{\ell}_{jt}^{Total,-g}$<br>(2) |
|------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------|
| ACQUIRER $\times$ POST                                     | 0.174*** (0.058)                                                 | 0.244*** (0.052)                    |
| ACQUIRER $\times$ POST $\times \widehat{\Delta L}^{Total}$ | 6.220*** (1.595)                                                 | 7.651** (2.874)                     |
| Controls                                                   | Yes                                                              | Yes                                 |
| Cohort-Bank FE                                             | Yes                                                              | Yes                                 |
| Cohort-Year FE                                             | Yes                                                              | Yes                                 |
| Observations                                               | 27,948                                                           | 27,948                              |
| Adjusted R <sup>2</sup>                                    | 0.878                                                            | 0.604                               |
| Average Y                                                  | 2.790                                                            | 0.179                               |

## 6 Conclusion

This paper studies how bank pricing power distorts the allocation of deposits and loans across banks. We develop a framework in which the welfare loss from these distortions is summarized by a set of sufficient statistics. The key objects are the dispersion of deposit and loan markups, weighted by a reallocation sensitivity matrix, together with an interaction term that captures the two-sided nature of banking.

Empirically, the measured welfare loss from markup-driven misallocation is sizable early in the sample and declines markedly over time. The deposit side accounts for most of the loss, and deposit and loan distortions usually reinforce each other in welfare. Counterfactuals also show that cross-side demand links matter for misallocation through two margins. They strengthen cross-side transmission in welfare, but they also compress markup dispersion within each side. In our estimates, the second force dominates, so shutting down the cross-side demand link raises the total welfare loss.

The merger results further show that consolidation has heterogeneous allocative effects. In the stacked event-study analysis, acquirers’ absolute social wedges and welfare loss con-

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measurement error. The best practice is to implement a first-stage propagation bootstrap.

tributions rise relative to those of control banks after the deal, with the strongest effect on the deposit side. In the *ex ante* exercise, most deals are close to neutral, but the distribution has a meaningful right tail of deals that worsen misallocation. Deals with worse *ex ante* predicted welfare loss also tend to worsen misallocation *ex post*.

These results suggest that evaluating competition in banking requires more than tracking average rates, spreads, or concentration. What also matters is how pricing wedges are distributed across banks, because that distribution governs where deposits and loans are allocated.

Our framework can be extended in several directions. A natural next step is to incorporate cost-side sources of misallocation in banking, such as size-based regulation and too-big-to-fail subsidies, and to embed banking-side misallocation in a richer equilibrium framework that also captures how these distortions affect the allocation of credit across firms and households. More broadly, the model can be adapted to study allocative efficiency in other two-sided markets. We leave these questions for future work.

## Appendix A Extra Results

### A.1 Decomposition of the Interaction Component

The interaction term is positive for most years, which implies that deposit and loan markups usually amplify the welfare loss from each other. A useful question is what drives that sign over time. One possibility is that the same banks tend to be socially too small or too large on both sides of the balance sheet, i.e., the deposit and loan social wedges move together. The other is that the cross-side block  $G^{DL}$  transmits a distortion on one side into a reinforcing effect on the other. Formally, the first force comes from the sign alignment of  $\omega^D$  and  $\omega^L$ , where each  $\omega$  measures a bank's markup relative to the common planner benchmark. The second force comes from the sign pattern of  $G^{DL}$ , which is shaped by demand elasticities together with the curvature of banks' balance-sheet costs, and therefore captures how easily deposit and loan distortions propagate across sides.

Recall that the welfare loss from the interaction term in Proposition 1 equals

$$L_X = \omega^{D\top} G^{DL} \omega^L = \sum_{j,k} \omega_j^D G_{jk}^{DL} \omega_k^L,$$

where  $(j, k)$  are a pair of bank  $j$  and  $k$ . We begin by separating the *absolute scale* of the interaction term from its *sign*. Write

$$L_X = \sum_{j,k} |\omega_j^D| |G_{jk}^{DL}| |\omega_k^L| \cdot \text{sign}(\omega_j^D \omega_k^L) \cdot \text{sign}(G_{jk}^{DL}),$$

where  $|\omega_j^D| |G_{jk}^{DL}| |\omega_k^L| \geq 0$  is the amount of interaction pressure attached to bank pair  $(j, k)$  before any positive and negative pieces offset each other. It is large when the deposit wedge is large, the loan wedge is large, and the cross-side linkage is strong. The sign term  $\text{sign}(\omega_j^D \omega_k^L)$  captures whether the social wedges on the two sides point in the same allocative direction. The sign term  $\text{sign}(G_{jk}^{DL})$  captures how much wedges on one side impact required quantity allocation on the other side through demand elasticities and balance-sheet cost convexities.

For ease of notation later, define

$$\begin{aligned}
L(\emptyset) &\equiv \sum_{j,k} |\omega_j^D| |G_{jk}^{DL}| |\omega_k^L|, \\
L(\omega) &\equiv \sum_{j,k} |\omega_j^D| |G_{jk}^{DL}| |\omega_k^L| \cdot \text{sign}(\omega_j^D \omega_k^L), \\
L(G) &\equiv \sum_{j,k} |\omega_j^D| |G_{jk}^{DL}| |\omega_k^L| \cdot \text{sign}(G_{jk}^{DL}), \\
L(\omega, G) = L_X &\equiv \sum_{j,k} |\omega_j^D| |G_{jk}^{DL}| |\omega_k^L| \cdot \text{sign}(\omega_j^D \omega_k^L) \cdot \text{sign}(G_{jk}^{DL}),
\end{aligned}$$

where  $L(\emptyset)$  is the all-positive benchmark, the hypothetical loss that would arise if every bank pair reinforced rather than offset the others. The object  $L(\omega)$  turns on only the sign pattern of  $(\omega^D, \omega^L)$ ,  $L(G)$  turns on only the sign pattern of  $G^{DL}$ , and  $L(\omega, G)$  is the realized interaction loss.

This leads to the interaction sign index

$$L_X^{Sign} = \frac{L_X}{L(\emptyset)} \in [-1, 1],$$

which measures how much of the all-positive benchmark survives once the actual sign patterns are imposed. It is a clean proxy that separates the sign component of  $L_X$  from its size. A value close to one means that the actual social wedges and the cross-side reallocation sensitivities mostly line up in a reinforcing way. A value near zero means that positive and negative cells largely cancel. A negative value means that, even though the absolute interaction pressure may still be sizable, the sign pattern is adverse enough that the net interaction turns negative.

The right panel of Figure 4 shows that  $L_X^{Sign}$  is around 0.3 in 1986, rises to roughly 0.6 by 2012, and turns slightly negative by 2021. The message is that most bank pairs do not contribute with the same intensity over time. Sometimes the social wedges and the reallocation sensitivity block line up so that two-sidedness amplifies the welfare loss. At other times they offset each other much more strongly.

To understand which sign channel matters more, we further decompose  $L_X^{Sign}$  into a social wedge ( $\omega$ ) sign index and a curvature ( $G^{DL}$ ) sign index. Because the two sign channels enter multiplicatively, a sequential decomposition is path-dependent. We therefore decompose the gap  $L_X - L(\emptyset)$  using the two-player Shapley value.

The contributions of the social wedge and the curvature sign channels are

$$\begin{aligned}\Lambda_\omega &= \frac{1}{2} \left( [L(\omega) - L(\emptyset)] + [L(\omega, G) - L(G)] \right), \\ \Lambda_G &= \frac{1}{2} \left( [L(G) - L(\emptyset)] + [L(\omega, G) - L(\omega)] \right).\end{aligned}$$

This decomposition is exact,

$$\begin{aligned}L_X &= L(\emptyset) + \Lambda_\omega + \Lambda_G, \\ \frac{L_X}{L(\emptyset)} &= 1 + \frac{\Lambda_\omega}{L(\emptyset)} + \frac{\Lambda_G}{L(\emptyset)} \quad \Rightarrow \quad L_X^{Sign} = 1 + L_\omega^{Sign} + L_G^{Sign}.\end{aligned}$$

It is also order-invariant, i.e., each channel receives its average marginal effect across the two possible orderings.

The two normalized Shapley terms have a direct economic interpretation. The term  $L_\omega^{Sign} \in [-2, 0]$  measures how much of the hypothetical all-positive benchmark is reduced because deposit-side and loan-side wedges are not aligned in the banks and bank pairs that carry the most weight<sup>16</sup>. When  $L_\omega^{Sign}$  is more negative, markups are less synchronized across the two sides of the balance sheet, i.e., banks that are socially too small on deposits are less likely to be the ones that are also socially too small on loans, and likewise for banks that are too large.

The term  $L_G^{Sign} \in [-2, 0]$  measures how much the hypothetical all-positive benchmark is decreased because of the sign pattern of  $G^{DL}$ . When it is more negative, the reallocation sensitivity of the banking system, coming from both demand linkages and cost convexity,

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<sup>16</sup>The Shapley contribution shares  $\frac{\Lambda_\omega}{L(\emptyset)}$  and  $\frac{\Lambda_G}{L(\emptyset)}$  are bounded by  $[-2, 0]$ . To see this, note that

$$\begin{aligned}\Lambda_\omega &= \frac{1}{2} \sum_{j,k} |\omega_j^D| |G_{jk}^{DL}| |\omega_k^L| \left[ (\text{sign}(\omega_j^D \omega_k^L) - 1) + \text{sign}(G_{jk}^{DL})(\text{sign}(\omega_j^D \omega_k^L) - 1) \right] \\ &= \frac{1}{2} \sum_{j,k} |\omega_j^D| |G_{jk}^{DL}| |\omega_k^L| (\text{sign}(\omega_j^D \omega_k^L) - 1)(1 + \text{sign}(G_{jk}^{DL})).\end{aligned}$$

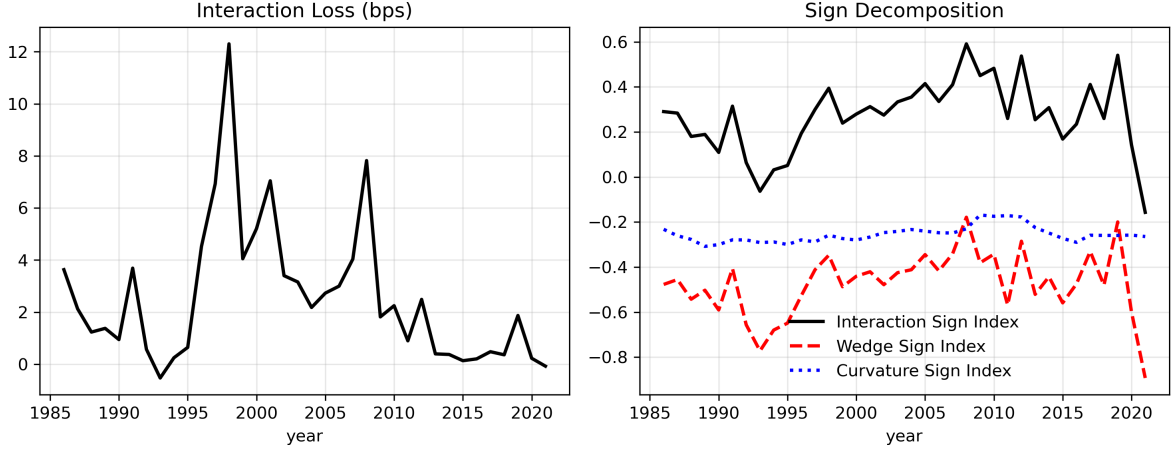
If  $\text{sign}(G_{jk}^{DL}) = -1$ , the term is zero. If  $\text{sign}(G_{jk}^{DL}) = 1$ , then the term is 0 when  $\text{sign}(\omega_j^D \omega_k^L) = 1$  and  $-2|\omega_j^D| |G_{jk}^{DL}| |\omega_k^L|$  when  $\text{sign}(\omega_j^D \omega_k^L) = -1$ . Hence

$$\Lambda_\omega = -2 \sum_{j,k: \text{sign}(G_{jk}^{DL})=1, \text{sign}(\omega_j^D \omega_k^L)=-1} |\omega_j^D| |G_{jk}^{DL}| |\omega_k^L|,$$

so  $\frac{\Lambda_\omega}{L(\emptyset)} \in [-2, 0]$ , and similarly  $\frac{\Lambda_G}{L(\emptyset)} \in [-2, 0]$ . Both Shapley shares are weakly negative, and each measures how much that sign channel pulls the interaction term below the all-positive benchmark.



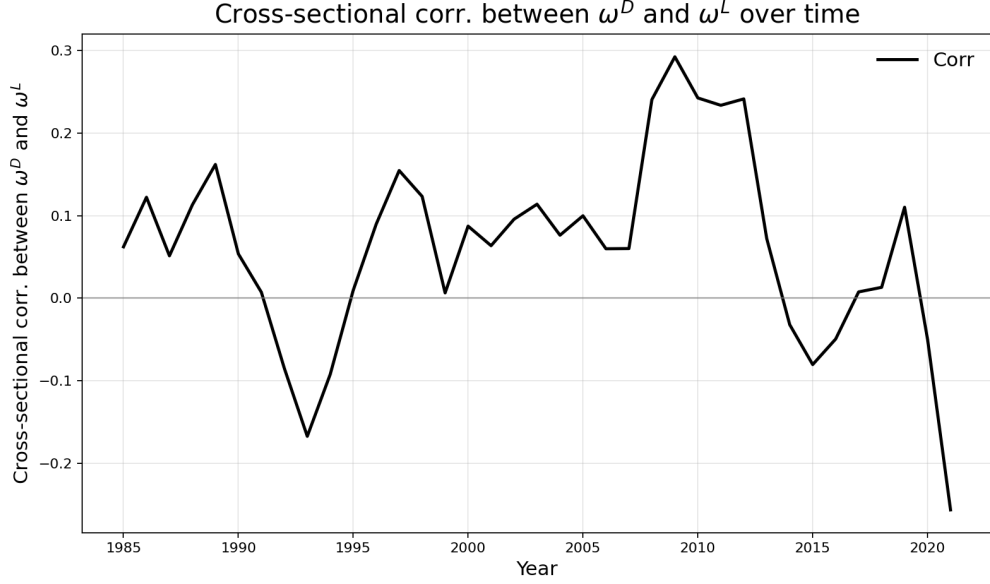
causes deposit-side and loan-side distortions to offset each other. Given demand complementarities and cost synergies within banks, the diagonal entries of  $G^{DL}$  are positive. By contrast, diversion across banks can make the off-diagonal entries negative, which in turn pushes  $L_G^{Sign}$  downward.



**Figure 4: Decomposition of the Interaction Component of Welfare Loss**

This figure plots the interaction component of the welfare loss and its sign decomposition over time. The black solid line is the normalized interaction component,  $L_X^{Sign}$ . The red dashed line is the contribution of the social wedge sign channel,  $L_\omega^{Sign}$ . The blue dotted line is the contribution of the curvature sign channel,  $L_G^{Sign}$ . The identity is  $1 + L_\omega^{Sign} + L_G^{Sign}$ .

Figure 4 shows that most of the movement in the interaction sign index comes from the social wedge channel rather than the curvature channel. The curvature sign index is fairly stable over time, which suggests a persistent but modest drag on the hypothetical baseline amplification. By contrast, the social wedge sign index varies substantially and explains most of the decline in the interaction sign index in later years. Figure 5 plots the cross-sectional correlation between the two wedges over time and confirms the variation in Figure 4. Economically, this pattern implies that what changes over time is mainly whether deposit and loan markups push misallocation in the same direction. The way demand complementarities and balance-sheet synergies transmit distortions across the two sides is much more stable. Thus, the social wedge channel is the main time-series driver of whether the two sides amplify or offset each other in welfare terms.



**Figure 5: Cross-Sectional Correlation between Deposit and Loan Wedges over Time**

This figure plots the cross-sectional Pearson correlation coefficient between deposit-side and loan-side social wedges ( $\omega^D, \omega^L$ ) over time.

## A.2 *Ex Ante* Misallocation Loss from Mergers

The key object in our *ex ante* application is a measure of predicted welfare loss for each merger deal, based on a counterfactual that changes only bank ownership. We deliberately shut down cost synergies or other merger-specific changes in technology, so that the exercise focuses on the *ex ante* misallocation consequences of *ownership-induced changes in markups*, not on merger efficiencies more broadly.

Consider a realized merger  $g = (j, k, T_g)$ , in which bank  $j$  acquires bank  $k$  at date  $T_g$ . We compute counterfactual markups under an alternative ownership matrix that treats  $j$  and  $k$  as jointly owned. Under pre-merger ownership, bank  $j$  views diversion from  $j$  to  $k$  as business lost to a rival. Under joint ownership, that diversion is internalized because the owner values profits from both  $j$  and  $k$ . The merger is therefore modeled as a change from single-bank ownership to multi-bank ownership, rather than a literal aggregation into one merged bank.

Recall from (17), the markups  $\mu_t \equiv (\mu_t^D, \mu_t^L)$  are jointly computed as

$$\mu_t = -(\Omega_t \odot \Delta_t)^{-1} Q_t.$$

where  $Q_t \equiv (D_t, L_t)$  denotes the stacked vector of observed quantities,  $\Delta_t = \begin{bmatrix} D_{-r^D,t} & L_{-r^D,t} \\ D_{r^L,t} & L_{r^L,t} \end{bmatrix}$  is the two-sided Jacobian, and  $\Omega_t$  is the  $2J \times 2J$  ownership matrix.

Our merger simulations modify the ownership  $\Omega$ . An ownership structure is summarized by a  $J \times J$  matrix  $O_J$  with entry  $O_{(j,k)} = 1$  if products  $j$  and  $k$  are jointly internalized by the same owner, and 0 otherwise. The corresponding two-sided ownership matrix is

$$\Omega(O) = \begin{bmatrix} O_J & O_J \\ O_J & O_J \end{bmatrix}. \quad (33)$$

In the baseline with one product per side for each bank and no multi-bank ownership,  $O^0 = I_J$ , so  $\Omega(O^0) = \begin{bmatrix} I_J & I_J \\ I_J & I_J \end{bmatrix}$ . Under a merger between banks  $j$  and  $k$ ,  $O^{jk}$  is identical except that the  $(j, k)$  and  $(k, j)$  entries are also set to one, which in turn adds the corresponding within-side and cross-side internalization terms in  $\Omega(O^{jk})$ .

Accordingly, the model-implied markup vector under ownership structure  $O$  is

$$\mu_t(O) = -(\Omega(O) \odot \Delta_t)^{-1} Q_t.$$

In particular, under the observed pre-merger ownership structure  $O^0$ ,

$$\mu_t^0 \equiv \mu_t(O^0),$$

and under the counterfactual merger between banks  $j$  and  $k$ ,

$$\mu_t^{jk} \equiv \mu_t(O^{jk}).$$

We then obtain the social wedge between banks' markups and the optimal benchmark  $(\omega_t^{D,0}, \omega_t^{D,jk})$  and  $(\omega_t^{L,0}, \omega_t^{L,jk})$ , using the same method introduced in Section 3.2.2. We then plug the wedges into the same local welfare map  $L_t(\cdot)$  from Proposition 1, *keeping the observed balance-sheet quantities and  $\Gamma_t$  at their original pre-merger level*, and obtain

$$L_t^0 \equiv L_t(\omega_t^0), \quad L_t^{jk} \equiv L_t(\omega_t^{jk}).$$

We define the predicted changes in welfare loss  $\Delta L$  for the merger of any two banks  $j$  and  $k$ , based on the *observed balance-sheet structure and reallocation sensitivity  $G$  estimates* in

year  $t$ , as

$$\Delta L_t^{jk} \equiv L_t^{jk} - L_t^0. \quad (34)$$

Finally, for each merger deal  $g = (j, k, T_g)$ , we define a pre-merger scoring window  $s \in \mathcal{H}_g$ . Based on the estimates and data within the window, we construct the *ex ante* predicted misallocation loss for merger deal  $g$ :

$$\Delta L_g = \mathcal{A}(\{\Delta L_{s \in \mathcal{H}_g}^{(j,k) \in g}\}), \quad (35)$$

where  $\mathcal{A}(\cdot)$  is an aggregator such as the mean. Averaging over a pre-merger window smooths transitory measurement noise in any single year.

$\Delta L_g$  is in the same dollar unit as the welfare loss. A positive  $\Delta L_g$  means that moving from the observed ownership structure to the merged ownership structure increases the welfare loss, while a negative  $\Delta L_g$  means that the merger reduces the measured misallocation. A merger is predicted to be welfare-improving in our sense when the ownership-induced change in markups compresses the planner-weighted dispersion of social wedges across banks.

Furthermore, the deal-level predicted loss inherits the decomposition in Proposition 1:

$$\Delta L_g = \Delta L_g^D + \Delta L_g^L + \Delta L_g^X, \quad (36)$$

where the three terms are the welfare changes in the deposit component, the loan component, and the interaction component, respectively. This decomposition is useful because a merger could worsen the dispersion of markups on one side in isolation and yet still lower total misallocation.

Because the level of  $\Delta L_g$  is hard to compare across years, we report the changes in basis points. For any component  $p \in \{\text{Total}, D, L, X\}$ , define

$$\widehat{\Delta L}_g^p \equiv 10000 \times \frac{\Delta L_g^p}{|L_g^{0, \text{Total}}|}, \quad (37)$$

where  $L_g^{0, \text{Total}}$  is the aggregated pre-merger total loss.  $\widehat{\Delta L}_g^p$  is the basis-point (bps) change in the relevant welfare loss, with negative values again indicating a reduction in markup-driven misallocation.

This construction is important because it isolates ownership from measurement. Both the

no-merger benchmark and the merger counterfactual are generated from the same ownership-based pricing system, using the same estimated two-sided demand Jacobian, the same observed bank-level quantities, and the same reallocation sensitivity matrix  $G$ . The predicted loss measure therefore answers a narrow but economically meaningful question: if ownership changes while the local allocation environment is held fixed, does the implied distribution of markups move closer to or farther from the planner’s benchmark?

**Table 8: Predicted Misallocation Loss for M&A Deals**

This table reports the distribution of the predicted welfare loss  $\widehat{\Delta L}_g$  for each merger deal  $g$ .  $\widehat{\Delta L}_g$  measures the deal-level basis point change in the welfare loss implied by ownership-induced repricing, holding fixed the pre-merger allocation, demand system, and welfare weights. Negative values indicate a reduction in misallocation. We report the deposit, loan, interaction, and total components separately.

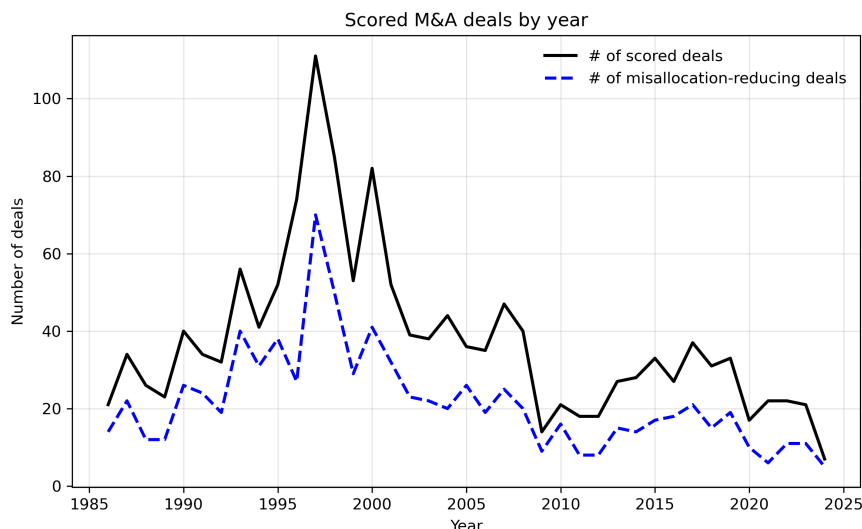
| Component   | Mean  | P10    | P25    | P50    | P75   | P90   | N    | Share negative |
|-------------|-------|--------|--------|--------|-------|-------|------|----------------|
| Total       | 0.254 | -0.315 | -0.079 | -0.007 | 0.064 | 0.685 | 1471 | 0.574          |
| Deposits    | 0.217 | -0.178 | -0.046 | -0.005 | 0.047 | 0.594 | 1471 | 0.579          |
| Loans       | 0.020 | -0.071 | -0.009 | 0.000  | 0.009 | 0.058 | 1471 | 0.475          |
| Interaction | 0.017 | -0.175 | -0.043 | -0.005 | 0.009 | 0.128 | 1471 | 0.633          |

Table 8 shows that most mergers have little effect on misallocation. The median total predicted welfare loss of a merger deal is  $-0.007$  bps, and the interquartile range is narrow, from  $-0.079$  to  $0.064$  bps relative to the pre-merger welfare loss benchmark. At the same time, the distribution is skewed to the right. While 57.4% of the deals reduce misallocation in the sense that their total predicted loss is negative, the mean predicted loss is positive at 0.254 bps. Overall, these results suggest that the typical deal is close to neutral or slightly misallocation-reducing, but a smaller group of misallocation-increasing mergers generates a sizable upper tail and drives the average effect above zero.

The decomposition in Table 8 helps explain this asymmetry. On the deposit side, 57.9% of mergers reduce deposit-side misallocation, yet the average predicted loss remains positive. By contrast, the loan and interaction components are both small on average and close to zero. In our sample, the deposit component is therefore the main driver of the overall merger effect on misallocation.

The predicted changes in misallocation are generally small. This is expected given the scale of the national banking market. Each counterfactual changes ownership for only the merging banks while holding fixed quantities, demand, and welfare weights. With hundreds of banks in a market, this ownership change perturbs only a small part of the pricing system, so the induced changes in markups are modest.

We further assess the time-series patterns on the extensive margin in Figure 6. The number of eligible deals rises rapidly from the late 1980s into the mid-1990s, peaks in the mid-1990s, declines around the early 2000s and the financial crisis, and then partly rebounds. The number of misallocation-reducing deals closely tracks the total number of eligible deals, which suggests that the results are not driven by a few isolated transactions. At the same time, the persistent gap between the two lines in every period reinforces the message from Table 8: even within the same merger wave, some deals move markups closer to the planner’s benchmark while others move them further away.



**Figure 6: Number of M&A Deals**

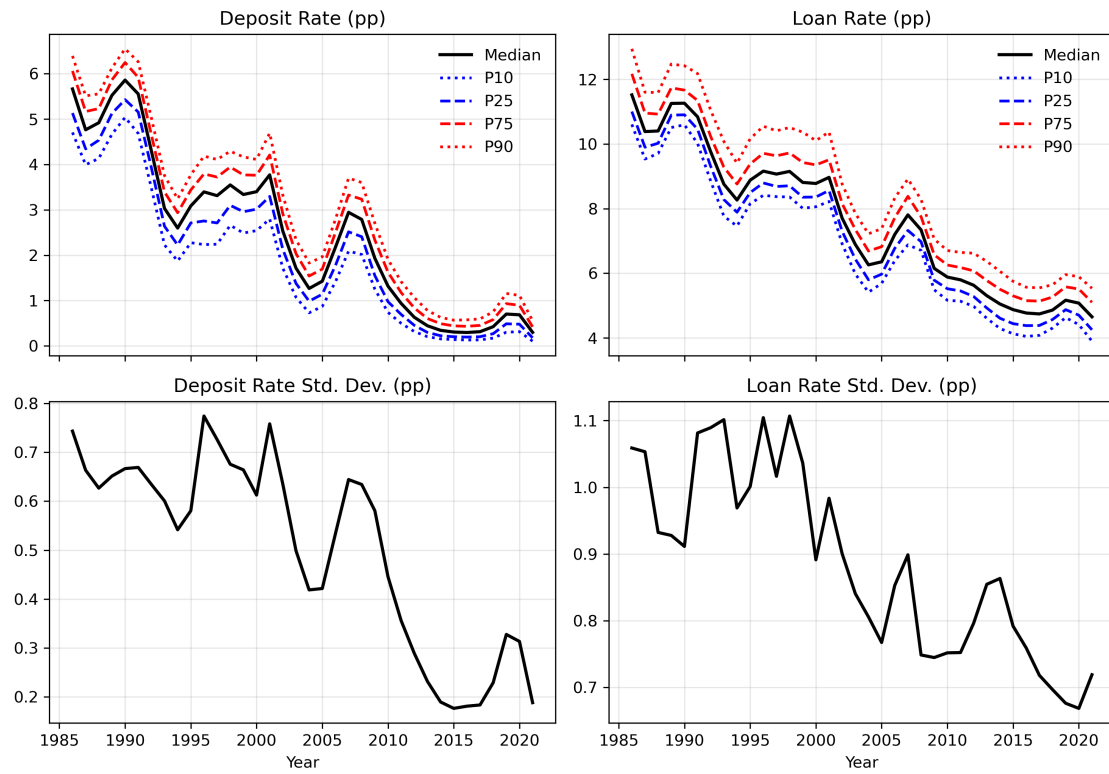
This figure plots the number of merger deals that receive a predicted loss from misallocation (black solid line), together with those classified as misallocation-reducing (blue dashed line).

Taken together, these results point to a nuanced view of consolidation. Our deal-level predicted loss measure does not suggest that mergers are uniformly harmful or uniformly beneficial from the perspective of two-sided misallocation. Instead, the typical deal is close to neutral, with a slight tendency toward reducing misallocation, but the distribution has a nontrivial right tail of deals that increase welfare loss. This is why an *ex ante* deal-level measure is useful. For policy, what matters most is not the average merger, but the ability to distinguish relatively benign consolidations from the smaller set of transactions that are predicted to worsen misallocation materially.

The *ex ante* exercise is deliberately partial equilibrium, which is both its strength and its limitation. We do not solve for a new post-merger quantity allocation, re-estimate demand, or allow reallocation sensitivity  $G$  to change after the deal. Nor do we incorporate cost

synergies, integration frictions, balance-sheet restructuring, branch closures, or other non-price channels. Thus, a negative  $\Delta L_g$  should not be read as a statement about total welfare in a global post-merger equilibrium. It means only that, within our local misallocation framework, merged ownership compresses markup-driven misallocation.

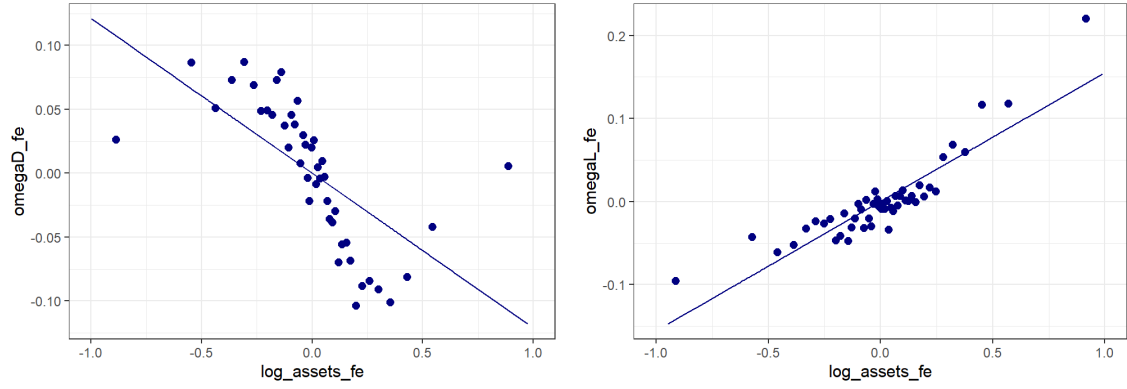
### A.3 More Tables and Figures



**Figure 7: Rates over Time**

This figure plots the cross-sectional dispersion of deposit and loan rates over time. All statistics are reported in percentage points.





**Figure 8: Wedges and Bank Size**

This figure shows binned scatter plots of social wedges ( $\omega^D, \omega^L$ ) against  $\log(\text{assets})$  after controlling for bank and year fixed effects.

**Table 9: Bank Mergers, Rates, and Markups**

This table reports estimates from the stacked regressions that relate banks' M&A activity to the absolute gap between markups and the social benchmark.  $ACQUIRER_{jc} = 1$  if bank  $j$  is an acquirer in cohort  $c$ , and equals 0 if bank  $j$  belongs to the control group in cohort  $c$ .  $POST_{ct} = 1$  in the deal year and in all subsequent years. The control variables include log assets, equity ratio, and the standard deviation of quarterly ROA in the past 2 years. Each is fixed at its pre-merger level at  $T_c - 3$  for each bank and then interacted with  $POST_{ct}$ . The standard errors in parentheses are clustered by bank and cohort. \*, \*\*, and \*\*\* denote significance at the 10%, 5%, and 1% level, respectively.

| Dependent Variables:   | $r^D$           | $r^L$            | $\hat{\mu}^D$ | $\hat{\mu}^L$    |
|------------------------|-----------------|------------------|---------------|------------------|
|                        | (1)             | (2)              | (3)           | (4)              |
| $ACQUIRER \times POST$ | 0.076** (0.032) | 0.163*** (0.049) | 0.014 (0.045) | 0.202*** (0.056) |
| Controls               | Yes             | Yes              | Yes           | Yes              |
| Cohort-Bank FE         | Yes             | Yes              | Yes           | Yes              |
| Cohort-Year FE         | Yes             | Yes              | Yes           | Yes              |
| Observations           | 27,948          | 27,948           | 27,948        | 27,948           |
| Adjusted $R^2$         | 0.978           | 0.966            | 0.777         | 0.795            |

## Appendix B Proofs

### B.1 Surplus

**Setup.** Let the (deterministic) part of depositor  $i$ 's utility from bank  $j$  be

$$V_{ij}^D \equiv -\alpha_i^D r_j^D + \beta^D X_j^D + \delta_j^D, \quad (38)$$

and the full indirect utility be

$$u_{ij}^D = V_{ij}^D + \epsilon_{ij}^D, \quad (39)$$

where  $\epsilon_{ij}^D$  are i.i.d. Type-I extreme value shocks (the standard logit case). Include an outside option  $j = 0$  with normalized utility  $V_{i0}^D = 0$ .

**Choice probabilities (within type).** Under i.i.d. Type-I EV shocks, the probability that type  $i$  chooses bank  $j$  is

$$s_{ij}^D = \frac{\exp(V_{ij}^D)}{1 + \sum_{n=1}^J \exp(V_{in}^D)}. \quad (40)$$

Aggregating over types with weights  $\nu_i^D$  yields the market share expression:

$$s_j^D = \sum_{i=1}^I \nu_i^D s_{ij}^D. \quad (41)$$

**Expected maximum utility (log-sum / inclusive value).** A key logit property is that the expected maximum utility over the  $J + 1$  options is

$$\mathbb{E}_\epsilon \left[ \max_{j \in \{0, 1, \dots, J\}} \{V_{ij}^D + \epsilon_{ij}^D\} \right] = \gamma + \log \left( \sum_{j=0}^J \exp(V_{ij}^D) \right), \quad (42)$$

where  $\gamma$  is the Euler–Mascheroni constant. With  $V_{i0}^D = 0$ , this becomes

$$\mathbb{E}_\epsilon \left[ \max_j u_{ij}^D \right] = \gamma + \log \left( 1 + \sum_{j=1}^J \exp(V_{ij}^D) \right). \quad (43)$$

The constant  $\gamma$  does not affect marginal welfare comparisons (it drops out of differences), so

it is common to omit it when reporting consumer surplus.

**Consumer surplus (rate-equivalent units).** If we want surplus in *rate-equivalent* units for type  $i$  (i.e., the compensating variation measured in units of  $r$ ), we divide expected max utility by  $\alpha_i^D$ :

$$CS_i^D \equiv \frac{1}{-\alpha_i^D} \log \left( 1 + \sum_{j=1}^J \exp(V_{ij}^D) \right) = \frac{1}{-\alpha_i^D} \log \left( 1 + \sum_{j=1}^J \exp(-\alpha_i^D r_j^D + \beta^D X_j^D + \delta_j^D) \right). \quad (44)$$

**Market-level surplus (mixing over types).** With type weights  $\nu_i^D$  and market size  $M^D$  (number of depositors), total market surplus is

$$CS^D \equiv M^D \sum_{i=1}^I \nu_i^D CS_i^D = M^D \sum_{i=1}^I \nu_i^D \frac{1}{-\alpha_i^D} \log \left( 1 + \sum_{j=1}^J \exp(-\alpha_i^D r_j^D + \beta^D X_j^D + \delta_j^D) \right), \quad (45)$$

which matches (4).

**Derivative with respect to an offered rate.** Differentiate type- $i$  surplus with respect to  $r_j^D$ :

$$\frac{\partial CS_i^D}{\partial r_j^D} = \frac{1}{-\alpha_i^D} \cdot \frac{1}{1 + \sum_{n=1}^J \exp(V_{in}^D)} \cdot \exp(V_{ij}^D) \cdot \frac{\partial V_{ij}^D}{\partial r_j^D} \quad (46)$$

$$= \frac{1}{-\alpha_i^D} \cdot \frac{\exp(V_{ij}^D)}{1 + \sum_{n=1}^J \exp(V_{in}^D)} \cdot (-\alpha_i^D) \quad (47)$$

$$= \frac{\exp(V_{ij}^D)}{1 + \sum_{n=1}^J \exp(V_{in}^D)} = s_{ij}^D. \quad (48)$$

Aggregating across types and multiplying by market size,

$$\frac{\partial CS^D}{\partial r_j^D} = M^D \sum_{i=1}^I \nu_i^D s_{ij}^D = M^D s_j^D = D_j \quad (49)$$

## B.2 SP's FOCs

To highlight the intuition, we first assume one-sided demand, where deposits and loans do not depend on each other. The planner's FOC with respect to the deposit rate is then

$$\begin{aligned}\frac{\partial TS}{\partial r_j^D} &= \frac{\sum_k \Pi_k}{\partial r_j^D} + \frac{\partial CS^D}{\partial r_j^D} = -D_j - \sum_k r_k^D \frac{\partial D_k}{\partial r_j^D} - \sum_k \frac{\partial C_k^{SP}}{\partial D_k} \frac{\partial D_k}{\partial r_j^D} + D_j \\ &= \sum_k \left( -r_k^D - \frac{\partial C_k^{SP}}{\partial D_k} \right) \frac{\partial D_k}{\partial r_j^D} \\ &= \lambda^D \sum_k \frac{\partial D_k}{\partial r_j^D} \quad \text{social planner's FOC,}\end{aligned}$$

which is equivalent to  $MSD_j = -r_j^D - \frac{\partial C_j^{SP}}{\partial D_j} = \lambda^D$  in (21) as long as  $\frac{\partial D_j}{\partial r_j^D} \neq 0$ . The key point is that  $-D_j$  from  $\frac{\partial \Pi_j}{\partial r_j^D}$  cancels out with  $D_j$  from  $\frac{\partial CS^D}{\partial r_j^D}$ , which is the pure transfer from banks to depositors when  $r_j^D$  rises.

Because deposits and loans are jointly determined under the two-sided demand system, a change in one rate generally affects *both* sides' quantities. Let

$$Q \equiv \begin{bmatrix} D \\ L \end{bmatrix}, \quad r \equiv \begin{bmatrix} -r^D \\ r^L \end{bmatrix}, \quad \mathcal{J}(r) \equiv \frac{\partial Q}{\partial r} = \begin{bmatrix} \frac{\partial D}{\partial (-r^D)} & \frac{\partial D}{\partial r^L} \\ \frac{\partial L}{\partial (-r^D)} & \frac{\partial L}{\partial r^L} \end{bmatrix}.$$

The planner's Lagrangian is

$$\mathcal{L}^{SP}(r, \lambda^D, \lambda^L) = TS(r) - \lambda^D (\mathbf{1}^\top D(r) - \bar{D}) - \lambda^L (\mathbf{1}^\top L(r) - \bar{L}).$$

Write the bank-level marginal surpluses as

$$MSD_j(Q) \equiv -\frac{\partial C_j^{SP}(Q_j)}{\partial D_j} - r_j^D, \quad MSL_j(Q) \equiv r_j^L - \frac{\partial C_j^{SP}(Q_j)}{\partial L_j}.$$

Then the gradient of total surplus with respect to quantities is

$$\nabla_Q TS(Q) = \begin{bmatrix} MSD(Q) \\ MSL(Q) \end{bmatrix}.$$

Hence the planner's FOCs with respect to the *rates* can be written compactly as

$$\begin{aligned} 0 &= \nabla_r \mathcal{L}^{SP}(r, \lambda^D, \lambda^L) \\ &= \mathcal{J}(r)^\top \left( \begin{bmatrix} MSD(Q) \\ MSL(Q) \end{bmatrix} - \begin{bmatrix} \lambda^D \mathbf{1} \\ \lambda^L \mathbf{1} \end{bmatrix} \right). \end{aligned}$$

Equivalently, for each deposit rate  $r_j^D$ ,

$$\begin{aligned} 0 &= \frac{\partial \mathcal{L}^{SP}}{\partial r_j^D} \\ &= \sum_{k=1}^J (MSD_k - \lambda^D) \frac{\partial D_k}{\partial r_j^D} + \sum_{k=1}^J (MSL_k - \lambda^L) \frac{\partial L_k}{\partial r_j^D}, \end{aligned}$$

and similarly, for each loan rate  $r_j^L$ ,

$$\begin{aligned} 0 &= \frac{\partial \mathcal{L}^{SP}}{\partial r_j^L} \\ &= \sum_{k=1}^J (MSD_k - \lambda^D) \frac{\partial D_k}{\partial r_j^L} + \sum_{k=1}^J (MSL_k - \lambda^L) \frac{\partial L_k}{\partial r_j^L}. \end{aligned}$$

These expressions make clear why the one-sided proof is no longer valid once cross-side demand feedback is present: changing  $r_j^D$  affects not only deposits but also loans through the off-diagonal Jacobian block  $L_{r^D}$ , and changing  $r_j^L$  affects not only loans but also deposits through  $\frac{\partial D}{\partial r^L}$ .

If  $\Delta(r^*)$  has full rank at the social optimum, the stacked first-order condition above implies

$$\begin{bmatrix} MSD(Q^*) \\ MSL(Q^*) \end{bmatrix} = \begin{bmatrix} \lambda^D \mathbf{1} \\ \lambda^L \mathbf{1} \end{bmatrix},$$

that is,

$$\begin{aligned} MSD_j(r_j^{D*}, Q_j^*) &= -\frac{\partial C_j^{SP}(Q_j^*)}{\partial D_j^*} - r_j^{D*} = \lambda^D, \\ MSL_j(r_j^{L*}, Q_j^*) &= r_j^{L*} - \frac{\partial C_j^{SP}(Q_j^*)}{\partial L_j^*} = \lambda^L, \end{aligned}$$

for all banks  $j$ .

Thus, the equalization result in the main text remains correct under two-sided demand, but the proof must be written in stacked form. In the special case without cross-side demand feedback,  $\frac{\partial D}{\partial r^L} = 0$  and  $L_{r^D} = 0$ , and the argument collapses to the usual one-sided envelope-style derivation side by side.

### B.3 Proposition 1

We prove equation (25) and the curvature formulas in Proposition 1. Throughout, let

$$Q \equiv (D, L) \in \mathbb{R}^{2J}, \quad p \equiv (-r^D, r^L) \in \mathbb{R}^{2J},$$

so the joint demand Jacobian is

$$\mathcal{J}(p) \equiv \frac{\partial Q}{\partial p} = \begin{bmatrix} \partial D / \partial (-r^D) & \partial D / \partial r^L \\ \partial L / \partial (-r^D) & \partial L / \partial r^L \end{bmatrix}.$$

By local invertibility of  $\mathcal{J}(p^*)$ , there exists a local inverse-demand map  $p(Q)$  with Jacobian  $\partial p / \partial Q$ . Define

$$S \equiv -\frac{\partial p}{\partial Q} = \begin{bmatrix} \frac{\partial r^D}{\partial D} & \frac{\partial r^D}{\partial L} \\ -\frac{\partial r^L}{\partial D} & -\frac{\partial r^L}{\partial L} \end{bmatrix},$$

all evaluated at  $Q^*$  or equivalently at  $p^*$ . The bank-level marginal surpluses in quantity space are

$$MSD_j(Q) \equiv -\frac{\partial C_j^{SP}(Q_j)}{\partial D_j} - r_j^D(Q), \tag{50}$$

$$MSL_j(Q) \equiv r_j^L(Q) - \frac{\partial C_j^{SP}(Q_j)}{\partial L_j}. \tag{51}$$

The social wedges are

$$\omega^D \equiv \mu^D - \lambda^D \mathbf{1}, \quad \omega^L \equiv \mu^L - \lambda^L \mathbf{1},$$

where  $(\lambda^D, \lambda^L)$  are the planner's multipliers at  $(D^*, L^*)$ .

**Step 1 (Price-space gradient).** Write total surplus as  $TS(Q(p))$ . Using the signed price vector  $p = (-r^D, r^L)$ , a marginal increase in  $p_j$  raises bank  $j$ 's direct payoff by exactly the same amount that it lowers the corresponding direct consumer surplus component, so the pure transfer term cancels. What remains is the effect through the induced quantity changes. Therefore,

$$\nabla_p TS(Q(p)) = \mathcal{J}(p)^\top \begin{bmatrix} MSD(Q) \\ MSL(Q) \end{bmatrix}.$$

Equivalently, for each signed price component  $p_m$ ,

$$\frac{\partial TS}{\partial p_m} = \sum_{k=1}^J MSD_k(Q) \frac{\partial D_k}{\partial p_m} + \sum_{k=1}^J MSL_k(Q) \frac{\partial L_k}{\partial p_m}.$$

This is the stacked price-space analogue of the usual one-sided envelope argument.

**Step 2 (Quantity-space gradient via chain rule).** Since  $p(Q)$  exists locally, the chain rule gives

$$\begin{aligned} \nabla_Q TS(Q) &= \left( \frac{\partial p}{\partial Q} \right)^\top \nabla_p TS(Q(p)) \\ &= \left( \frac{\partial p}{\partial Q} \right)^\top \mathcal{J}(p)^\top \begin{bmatrix} MSD(Q) \\ MSL(Q) \end{bmatrix}. \end{aligned} \tag{52}$$

Because  $Q \mapsto p(Q)$  is the local inverse of  $p \mapsto Q(p)$ ,

$$\mathcal{J}(p) \frac{\partial p}{\partial Q} = I, \quad \left( \frac{\partial p}{\partial Q} \right)^\top \mathcal{J}(p)^\top = I.$$

Hence (52) simplifies to

$$\nabla_Q TS(Q) = \begin{bmatrix} MSD(Q) \\ MSL(Q) \end{bmatrix}. \tag{53}$$

**Step 3 (Planner's first-order conditions).** The planner's Lagrangian in quantity space is

$$\mathcal{L}(Q) = TS(Q) - \lambda^D (\mathbf{1}^\top D - \bar{D}) - \lambda^L (\mathbf{1}^\top L - \bar{L}).$$



Using (53), the first-order conditions at  $Q^*$  are

$$\nabla_Q \mathcal{L}(Q^*) = \begin{bmatrix} MSD(Q^*) - \lambda^D \mathbf{1} \\ MS L(Q^*) - \lambda^L \mathbf{1} \end{bmatrix} = 0.$$

Thus the social optimum equalizes marginal surplus across banks on each side, which is exactly the equalization result in (21)–(22). Define the stacked wedge

$$\omega(Q) \equiv \nabla_Q \mathcal{L}(Q) = \begin{bmatrix} \omega^D(Q) \\ \omega^L(Q) \end{bmatrix} = \begin{bmatrix} MSD(Q) - \lambda^D \mathbf{1} \\ MS L(Q) - \lambda^L \mathbf{1} \end{bmatrix}. \quad (54)$$

At the social optimum,  $\omega(Q^*) = 0$ .

**Step 4 (Second-order Taylor expansion).** Let  $\delta \equiv Q^{DE} - Q^*$  be the quantity gap between the decentralized equilibrium and the social optimum. Since total surplus is strictly concave near  $Q^*$ , define

$$\Gamma \equiv -\nabla_Q^2 TS(Q^*) \succ 0, \quad G \equiv \Gamma^{-1}.$$

A second-order Taylor expansion of  $\mathcal{L}$  around  $Q^*$  gives

$$\mathcal{L}(Q^{DE}) - \mathcal{L}(Q^*) = -\frac{1}{2} \delta^\top \Gamma \delta + o(\|\delta\|^2). \quad (55)$$

A first-order expansion of the gradient yields

$$\omega(Q^{DE}) = \nabla_Q^2 TS(Q^*) \delta + o(\|\delta\|) = -\Gamma \delta + o(\|\delta\|), \quad (56)$$

so locally  $\delta \approx -G\omega$ . Since both  $Q^{DE}$  and  $Q^*$  satisfy the aggregate constraints, the constraint terms in  $\mathcal{L}$  are the same at both points, and therefore

$$\mathcal{L}(Q^{DE}) - \mathcal{L}(Q^*) = TS(Q^{DE}) - TS(Q^*).$$

Substituting  $\delta \approx -G\omega$  into (55) gives

$$TS(Q^{DE}) - TS(Q^*) \approx -\frac{1}{2} \omega^\top G \omega.$$

Hence

$$TS^* - TS \approx \frac{1}{2}\omega^\top G\omega.$$

Partitioning

$$\Gamma = \begin{bmatrix} \Gamma_{DD} & \Gamma_{DL} \\ \Gamma_{LD} & \Gamma_{LL} \end{bmatrix}, \quad G = \begin{bmatrix} G_{DD} & G_{DL} \\ G_{LD} & G_{LL} \end{bmatrix}, \quad \omega = \begin{bmatrix} \omega^D \\ \omega^L \end{bmatrix},$$

we obtain

$$\omega^\top G\omega = \omega^{D\top} G_{DD}\omega^D + \omega^{L\top} G_{LL}\omega^L + 2\omega^{D\top} G_{DL}\omega^L,$$

and therefore

$$TS^* - TS \approx \frac{1}{2}\omega^{D\top} G_{DD}\omega^D + \frac{1}{2}\omega^{L\top} G_{LL}\omega^L + \omega^{D\top} G_{DL}\omega^L,$$

which is equation (25).

**Curvature blocks.** Differentiating (53) with respect to  $Q$  at  $Q^*$ ,

$$\Gamma = -\nabla_Q^2 TS(Q^*) = -\frac{\partial}{\partial Q} \begin{bmatrix} MSD(Q) \\ MS_L(Q) \end{bmatrix}_{Q=Q^*}. \quad (57)$$

Using (50)–(51) and  $\partial p/\partial Q = -S$ , the four blocks are

$$\Gamma_{DD} = C_{DD}^{SP} + S_{DD}, \quad \Gamma_{DL} = C_{DL}^{SP} + S_{DL},$$

$$\Gamma_{LD} = C_{LD}^{SP} + S_{LD}, \quad \Gamma_{LL} = C_{LL}^{SP} + S_{LL},$$

all evaluated at  $Q^*$ , with  $\Gamma_{LD} = \Gamma_{DL}^\top$  by symmetry of the Hessian. □

## B.4 Estimate Socially Optimal Markups

Let  $\mu_t \equiv (\mu_t^D, \mu_t^L) \in \mathbb{R}^{2J_t}$  stack markups and let

$$G_t \equiv \Gamma_t^{-1} = \begin{bmatrix} G_{DD,t} & G_{DL,t} \\ G_{LD,t} & G_{LL,t} \end{bmatrix}$$

stack the corresponding weight matrices in Proposition 1. Define the constraint-loading matrix

$$A \equiv \begin{bmatrix} \mathbf{1} & 0 \\ 0 & \mathbf{1} \end{bmatrix}, \quad \lambda_t \equiv \begin{bmatrix} \lambda_t^D \\ \lambda_t^L \end{bmatrix}.$$

Then the joint dispersion vector is  $\omega_t(\lambda_t) = \mu_t - A\lambda_t$  and the local reallocation is

$$\Delta Q_t \approx G_t \omega_t(\lambda_t), \quad Q_t \equiv (D_t, L_t).$$

The planner's aggregate constraints require both  $\mathbf{1}^\top \Delta D_t = 0$  and  $\mathbf{1}^\top \Delta L_t = 0$ , which compactly read

$$A^\top \Delta Q_t = 0.$$

Imposing this on the local reallocation implies a  $2 \times 2$  linear system for  $\lambda_t$ :

$$A^\top G_t (\mu_t - A\lambda_t) = 0 \implies \lambda_t = (A^\top G_t A)^{-1} A^\top G_t \mu_t.$$

Hence  $\lambda_t^D$  depends on both  $\mu_t^D$  and  $\mu_t^L$  whenever  $G_{DL,t} \neq 0$ , and similarly for  $\lambda_t^L$ .

This local method is consistent with Proposition 1. Proposition 1 is derived from a second-order expansion of total surplus around the planner's *constrained* optimum, where the multipliers  $\lambda^D, \lambda^L$  are precisely the shadow values that enforce the aggregate constraints. The reduced-form procedures above recover  $\lambda_t^D$  and  $\lambda_t^L$  by enforcing those same constraints *within the same local-quadratic approximation that links reallocations to social wedges through*  $G_t = \Gamma_t^{-1}$ . Consequently, the resulting  $\omega_t^D, \omega_t^L$  are the empirically implementable counterparts of the dispersion objects in Proposition 1.

## B.5 The Approximation Accuracy of the Reallocation Sensitivity Matrix

**Smoothness of curvature.** If  $TS$  is three-times continuously differentiable in a neighborhood of  $Q^*$ , then

$$\Gamma(Q) = \Gamma(Q^*) + O(\|Q - Q^*\|).$$

By standard matrix perturbation, this implies

$$G(Q) = \Gamma(Q)^{-1} = G(Q^*) + O(\|Q - Q^*\|).$$

**Wedges are first-order in distance to the optimum.** Define the KKT residual (social wedge) as  $\omega(Q) \equiv \nabla TS(Q) - A\lambda$ , so that  $\omega(Q^*) = 0$ . Then

$$\omega(Q) = O(\|Q - Q^*\|).$$

**Welfare loss is second-order, and the plug-in error is third-order.** The quadratic welfare approximation has the form

$$TS(Q) - TS(Q^*) \approx -\frac{1}{2} \omega(Q)^\top G(Q^*) \omega(Q).$$

Replacing  $G(Q^*)$  by  $G(Q)$  changes the quadratic form by

$$\omega(Q)^\top (G(Q) - G(Q^*)) \omega(Q) = O(\|Q - Q^*\|) \cdot O(\|Q - Q^*\|^2) = O(\|Q - Q^*\|^3).$$

Hence using  $\Gamma(Q)$  (or  $G(Q)$ ) evaluated at observed  $Q$  preserves the *second-order* accuracy of Proposition 1 up to a *third-order* remainder.

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